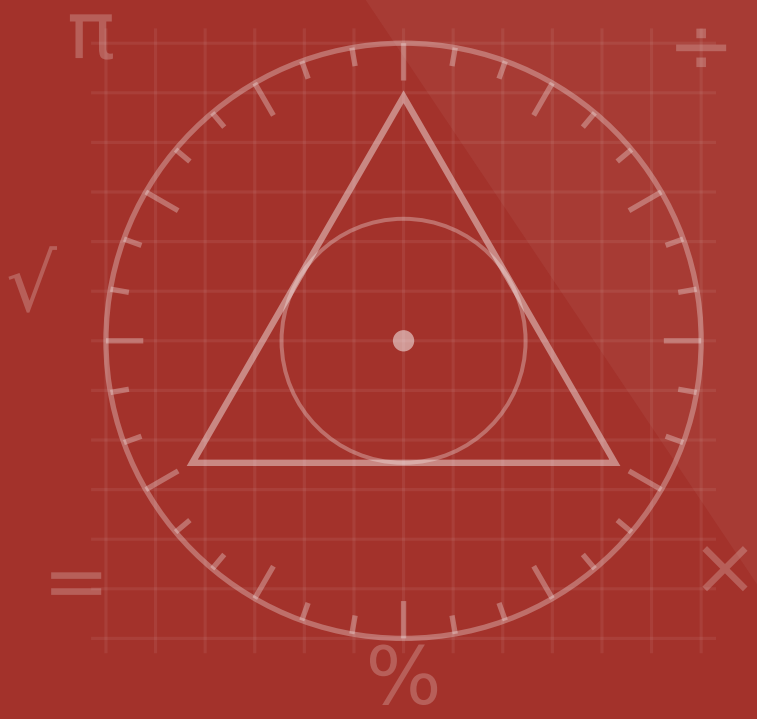
VENVLEARNING

STUDENT WORKBOOK



Mathematics

GRADE 7

Daily practice, weekly checks, and unit tests

2026–2027 School Year · 9 units

VENVLEARNING

Grade 7
Mathematics

STUDENT WORKBOOK

2026–2027 School Year

Unit 1 Proportional Relationships
Unit 2 Percents
Unit 3 Adding and Subtracting Integers
Unit 4 Adding and Subtracting Rational Numbers
Unit 5 Multiplying and Dividing Rational Numbers
Unit 6 Expressions, Equations, and Inequalities
Unit 7 Statistics and Probability
Unit 8 Scale Drawings and Geometric Figures
Unit 9 Angles, Circles, Area, and Volume

This book belongs to _____

A VenvLearning family homeschool curriculum

How to use this workbook

Every lesson in your textbook has matching practice pages here. Show your thinking in the work space. When a problem has an answer line, write your final answer there.

- Warm-Up reviews something you already know.
- Practice uses today's lesson.
- Think Deeper problems ask you to explain or find a mistake.
- Exit Ticket shows your teacher what you learned today.

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UNIT 1

Proportional Relationships

Ratios, unit rates, the constant of proportionality, and proportional tables, graphs, and equations.

1

Your practice pages for Unit 1. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

ratio	A comparison of two quantities.
unit rate	Amount per one unit.
proportional relationship	$y \div x$ is always the same.
constant of proportionality	$k = y \div x$.

UNIT 1 · LESSON 1.1 · PRACTICE

Lesson 1.1: Ratios and Rates

Name _____ Date _____

I can write ratios and rates and explain what they mean.

Warm-Up

1. 12 × 4

2. Simplify $\frac{18}{24}$

Write two equivalent ratios

3. 2:5

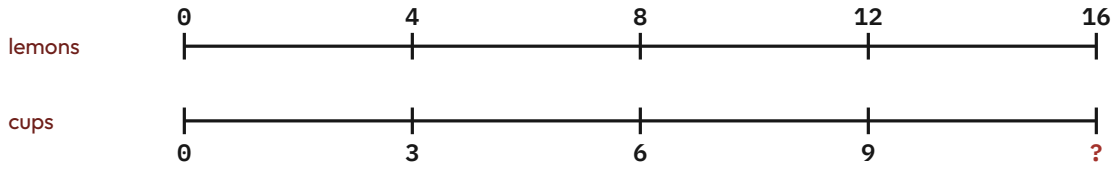
4. 7:3

5. 12:16

6. 1:4

Complete the double number line

7. Lemonade: 4 lemons for every 3 cups of water. How much water for 16 lemons?



Solve

8. A team won 12 games and lost 8. What is the ratio of wins to games played?

9. A printer prints 45 pages in 3 minutes. How many pages in 10 minutes?

Exit Ticket

10. Explain why 6:8 and 9:12 are equivalent.

UNIT 1 · LESSON 1.2 · PRACTICE

Lesson 1.2: Unit Rates

Name _____ Date _____

I can find a unit rate and use it to solve problems.

Warm-Up

1. 4.20 ÷ 12

2. 54 ÷ 6

Find the unit rate

3. $7.50 for 3 pounds

4. 120 km in 2.5 hours

5. 96 heartbeats in 1.5 minutes

6. $13.44 for 16 ounces

7. 8 laps in 20 minutes

8. $52 for 8 hours of work

Solve

9. Lillie earns $11.50 per hour babysitting. How much for 6 hours?

10. A car uses 3 gallons per 84 miles. How far on 10 gallons?

Exit Ticket

11. A 5-pound bag of rice costs $6.25. What is the price per pound? How much would 12 pounds cost?

UNIT 1 · LESSON 1.3 · PRACTICE

Lesson 1.3: Unit Rates with Fractions

Name _____ Date _____

I can find unit rates when the quantities are fractions, like $\frac{1}{2}$ mile in $\frac{1}{4}$ hour.

Warm-Up

1. $\frac{1}{2} \div \frac{1}{4}$

2. $\frac{2}{3} \times 3$

Find the unit rate

3. $\frac{1}{4}$ mile in $\frac{1}{8}$ hour

4. $\frac{2}{5}$ pound for $\frac{1}{5}$ dollar

5. $\frac{3}{4}$ page in $\frac{1}{4}$ minute

6. $\frac{5}{6}$ cup for $\frac{2}{3}$ batch

7. $1\frac{1}{2}$ miles in $\frac{1}{2}$ hour

8. $\frac{3}{8}$ gallon for $\frac{3}{4}$ acre

Solve

9. A snail moves $\frac{1}{10}$ meter in $\frac{1}{5}$ minute. How far in 1 minute? In 30 minutes?

Exit Ticket

10. Lillie mows $\frac{3}{5}$ of a lawn in $\frac{3}{4}$ hour. How much lawn per hour?

UNIT 1 · LESSON 1.4 · PRACTICE

Lesson 1.4: Comparing Unit Rates

Name _____ Date _____

I can use unit rates to decide which deal or option is better.

Warm-Up

1. 3.84 ÷ 16

2. 5.28 ÷ 24

Which is the better buy?

3. 8 oz for $2.40 or 12 oz for $3.24

4. 3 lb for $4.50 or 5 lb for $7.25

5. 6 pencils for $1.50 or 10 for $2.30

6. 2 L for $1.80 or 3 L for $2.85

Solve

7. Dad's truck goes 342 miles on 18 gallons. Mom's car goes 406 miles on 14 gallons. Whose vehicle gets better gas mileage, and by how many mpg?

Exit Ticket

8. Explain why a bigger package is not always the better buy.

WEEKLY CHECK · Fri Nov 6 · ABOUT 20 MINUTES

Weekly Check 1: Ratios and Unit Rates

Name _____ Date _____

Show what you know

1. Write two ratios equivalent to 4:9.

2. $11.40 for 6 pounds: unit price

3. $2/3$ mile in $1/6$ hour: mph

4. Better buy: 10 oz for $2.50 or 16 oz for $3.84

5. A car goes 252 miles on 9 gallons. mpg?

6. At 28 mpg, how far on 13 gallons?

UNIT 1 · LESSON 1.5 · PRACTICE

Lesson 1.5: What Is a Proportional Relationship?

Name _____ Date _____

I can tell when two quantities are in a proportional relationship.

Warm-Up

1. 21 ÷ 6

2. Is 4:10 equivalent to 6:15?

Proportional or not? Explain.

3. x: 1, 2, 3, 4 → y: 5, 10, 15, 20

4. x: 1, 2, 3, 4 → y: 3, 5, 7, 9

5. x: 2, 5, 8 → y: 3, 7.5, 12

6. x: 3, 6, 9 → y: 1, 2, 4

Real situations

7. Pay of $12 per hour

8. A taxi: $3 plus $2 per mile

9. Area of a square vs its side length

10. Cost of apples at $1.50 per pound

Exit Ticket

11. Make a table that is NOT proportional. Explain why.

UNIT 1 · LESSON 1.6 · PRACTICE

Lesson 1.6: The Constant of Proportionality

Name _____ Date _____

I can find the constant of proportionality k from a table.

Warm-Up

1. 7.5 ÷ 6

2. 45 ÷ 10

Find the constant of proportionality

3. x: 1, 3, 5 → y: 7, 21, 35

4. x: 2, 4, 10 → y: 3, 6, 15

5. x: 5, 10, 20 → y: 2, 4, 8

6. x: 6, 9, 12 → y: 4, 6, 8

Complete the proportional table

7. x: 1, 2, 5, 8 → y: 6, ___, ___, ___

8. x: 4, 10, 15 → y: 10, ___, ___

Exit Ticket

9. A car travels 165 miles in 3 hours at a steady speed. What is k? What does it mean?

UNIT 1 · LESSON 1.7 · PRACTICE

Lesson 1.7: Equations for Proportional Relationships

Name _____ Date _____

I can write an equation y = kx for a proportional relationship.

Warm-Up

1. 16 ÷ 2

2. 3.5 × 6

Write the equation

3. k = 9

4. x: 4, 8 → y: 10, 20

5. $0.65 per ounce

6. 15 pages per hour

Use the equation

7. y = 6x; x = 12

8. y = 0.8x; x = 25

9. y = 4x; y = 52, find x

10. $y = \frac{3}{4}x$; x = 20

Solve

11. Lillie earns $11.50 per hour. Write an equation. How much for 14 hours?

Exit Ticket

12. Explain what k means in y = 45x if x is hours and y is miles.

UNIT 1 · LESSON 1.8 · PRACTICE

Lesson 1.8: Is It Proportional? Tables and Equations

Name _____ Date _____

I can decide if a table or equation represents a proportional relationship.

Warm-Up

1. y = 7x when x = 0

2. y = 2x − 1 when x = 0

Proportional or not?

3. y = 9x

4. y = x + 9

5. $y = \frac{1}{3}x$

6. y = 3x + 0

7. y = 10 − x

8. y = 2.4x

Tables

9. x: 1, 2, 3 → y: 4, 8, 13

10. x: 0, 5, 10 → y: 0, 3, 6

Exit Ticket

11. A phone plan costs $25 plus $10 per GB. Is the cost proportional to GB? Explain.

WEEKLY CHECK · Fri Nov 13 · ABOUT 20 MINUTES

Weekly Check 2: Proportional Tables and Equations

Name _____ Date _____

Show what you know

1. x: 2, 5, 9 → y: 7, 17.5, 31.5. Proportional? k?

2. x: 1, 2, 3 → y: 5, 9, 13. Proportional?

3. Write the equation for k = 0.6.

4. y = 15x; x = 8

5. y = 4x; y = 30, find x

6. Is y = 3x + 2 proportional? Why?

UNIT 1 · LESSON 1.9 · PRACTICE

Lesson 1.9: Graphing Proportional Relationships

Name _____ Date _____

I can graph a proportional relationship on a coordinate plane.

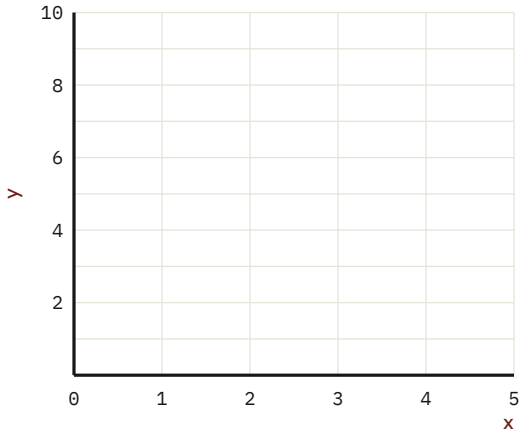
Warm-Up

1. y = 5x when x = 6

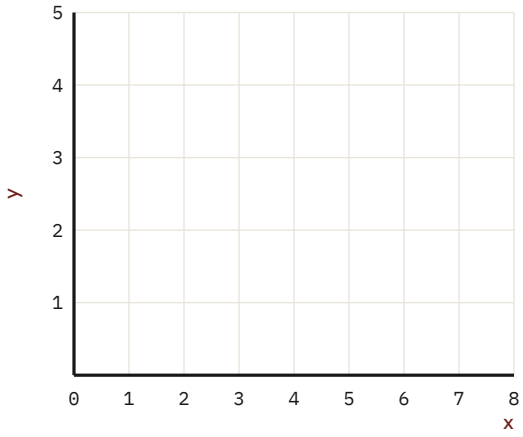
2. Name the point 4 right, 20 up

Make a table and graph

3. y = 2x for x = 0 to 5



4. y = 0.5x for x = 0 to 8



Exit Ticket

5. What do all proportional graphs have in common?

UNIT 1 · LESSON 1.10 · PRACTICE

Lesson 1.10: Recognizing Proportional Graphs

Name _____ Date _____

I can tell from a graph whether a relationship is proportional.

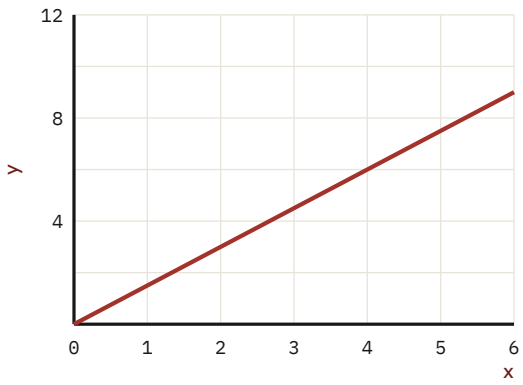
Warm-Up

1. 12 ÷ 3

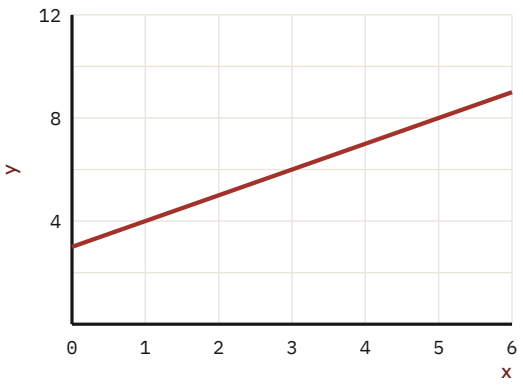
2. Is (0, 0) the origin?

Proportional or not?

3.



4.



Points on a line

5. A proportional line goes through (4, 10). Name two more points on it.

6. Does a line through (2, 6) and (5, 15) show a proportional relationship?

Exit Ticket

7. Sketch a graph that is a straight line but not proportional.

UNIT 1 · LESSON 1.11 · PRACTICE

Lesson 1.11: Interpreting Points on Proportional Graphs

Name _____ Date _____

I can explain what a point (x, y) means on a proportional graph, including (0, 0) and (1, k).

Warm-Up

1 28 × 3

2 84 ÷ 3

Interpret

3 A graph of cost vs pounds goes through (1, 4.5). What does it mean?

4 A graph of distance vs hours goes through (4, 260). Find k and explain.

5 A graph of cups of flour vs batches goes through (2, 5). What is the unit rate?

Exit Ticket

6 Write a story for a proportional graph through (1, 15).

UNIT 1 · LESSON 1.12 · PRACTICE

Lesson 1.12: Comparing Proportional Relationships

Name _____ Date _____

I can compare proportional relationships shown as tables, graphs, equations, and words.

Warm-Up

1. 30 ÷ 3

2. 18 ÷ 5

Compare

3. Pump 1: 12 gallons per minute. Pump 2: y = 14x. Faster?

4. Runner A: table 3 mi in 24 min. Runner B: y = 7.5x (min per mile). Faster?

5. Recipe A: 2 cups sugar per 5 cups flour. Recipe B: y = 0.35x (sugar per flour). Sweeter?

6. Plan A: $45 for 3 months. Plan B: graph through (1, 16). Cheaper?

Exit Ticket

7. Explain how to compare a table and a graph that both show proportional relationships.

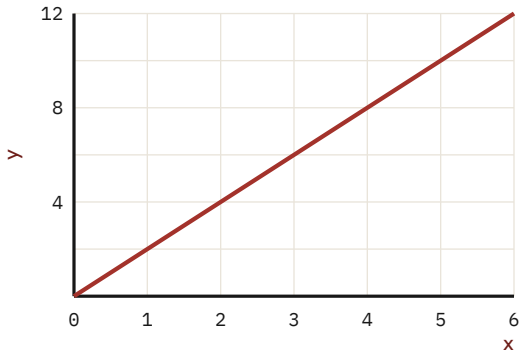
WEEKLY CHECK · Fri Nov 20 · ABOUT 20 MINUTES

Weekly Check 3: Proportional Graphs

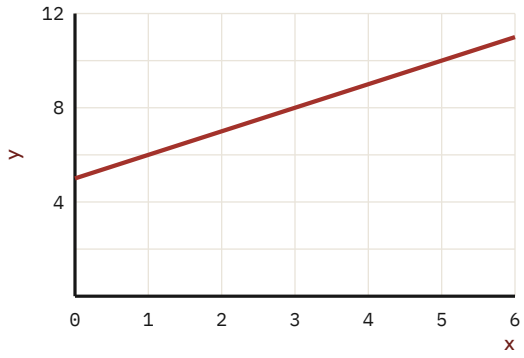
Name _____ Date _____

Show what you know

1 Proportional?



2 Proportional?



3 A proportional graph goes through (1, 6.5). What is k?

4 A proportional graph goes through (4, 18). What is y at x = 10?

5 Compare: y = 11x vs table 5 → 57.5. Greater rate?

6 What does (0, 0) mean on a graph of hours vs pay?

UNIT 1 · LESSON 1.13 · PRACTICE

Lesson 1.13: Solving Proportions

Name _____ Date _____

I can solve problems with proportions using unit rates, scale factors, or equations.

Warm-Up

1. $8.75 \div 5$

2. $96 \div 24$

Solve

3. $\frac{3}{8} = \frac{x}{40}$

4. $\frac{7}{x} = \frac{21}{33}$

5. $\frac{x}{12} = \frac{5}{4}$

6. $\frac{2.5}{4} = \frac{x}{10}$

Word problems

7. 6 bags of mulch cover 72 sq ft. How many bags for 300 sq ft?

8. A 20-oz bottle has 240 calories. How many calories in 12 oz?

9. Ellie reads 45 pages in 1.5 hours. How long for 120 pages?

10. Paint: 2 gallons per 700 sq ft. Gallons for 1,750 sq ft?

Exit Ticket

11. Solve two ways: 8 lemons make 3 quarts. How many lemons for 12 quarts?

UNIT 1 · LESSON 1.14 · PRACTICE

Lesson 1.14: Proportional Reasoning in the Real World

Name _____ Date _____

I can solve real-world problems with proportional relationships, including recipes, maps, and speed.

Warm-Up

1 $3/2 \div 6$

2 3.6×25

Solve

3 A recipe for 4 people uses 2 lb of chicken. How much for 14 people?

4 A cyclist rides 18 miles in 1.5 hours. At that rate, how far in 4 hours?

5 A model car is built at a scale of 1 in. : 18 in. The model is 9 in. long. How long is the real car in feet?

6 Lillie earns $57 for 6 hours of work. How many hours to earn $133?

Exit Ticket

7 Write and solve a recipe-scaling problem for your family.

UNIT 1 · LESSON 1.15 · PRACTICE

Lesson 1.15: Multi-Step Ratio Problems

Name _____ Date _____

I can solve multi-step problems involving ratios and rates.

Warm-Up

1. 20 ÷ 5 × 3

2. 100 + 195

Solve

3. A bag has red and green marbles in a ratio of 5:3. There are 48 marbles. How many are red?

4. A drink mix uses 2 scoops per 3 cups of water. How many scoops for a 9-quart cooler? (1 qt = 4 cups)

5. Lillie saves $3 for every $5 she earns. She earned $85. How much did she save? Spend?

6. A train travels 180 miles in 3 hours, then 120 miles in 1 hour. Average speed for the whole trip?

Exit Ticket

7. A ratio of cats to dogs at a shelter is 2:7. There are 63 animals. How many more dogs than cats?

UNIT 1 · LESSON 1.16 · PRACTICE

Lesson 1.16: Proportional Relationships Problem Solving

Name _____ Date _____

I can choose the best representation to solve proportional relationship problems.

Warm-Up

1. 5.50 ÷ 50

2. 0.12 × 100

Mixed practice

3. $\frac{1}{2}$ mile in $\frac{1}{8}$ hour. mph?

4. Better buy: 15 oz for $3.30 or 20 oz for $4.20?

5. x: 2, 3, 7 → y: 5, 7.5, 17.5. Equation?

6. A graph passes through (3, 7). Proportional? Need more information?

7. Ratio 3:4, total 56. Larger part?

8. $42 for 3.5 hours. Hourly rate?

Exit Ticket

9. Which representation do you find most useful for proportional relationships? Why?

UNIT REVIEW · Thu Dec 3 (homework) or as needed · ABOUT 40 MINUTES

Unit 1 Review

Name _____ Date _____

Rates and proportions

1. $9.60 for 12 cans: unit price

2. $\frac{3}{4}$ pound for $\frac{1}{2}$ dollar: pounds per dollar

3. Solve $\frac{9}{12} = \frac{x}{20}$

4. Better buy: 6 for $4.50 or 10 for $7.20

Proportional relationships

5. x: 3, 6, 12 → y: 2, 4, 8. Equation?

6. Is $y = 5x - 5$ proportional?

7. A proportional graph passes through (1, 3.25). Write the equation.

8. Ratio 2:5; total 63. Smaller part?

Solve

9. A recipe for 8 people uses 3 cups of broth. How much for 20 people?

UNIT TEST · Fri Dec 4 · ABOUT 45 MINUTES

Unit 1 Test

Name _____ Date _____

Show your work. This test has 20 points.

Part A · Rates (1 point each)

1. $13.75 for 5 pounds: unit price

2. $1/2$ cup per $2/3$ batch: cups per batch

3. Better buy: 24 oz for $4.08 or 32 oz for $5.12

4. Write two ratios equivalent to 6:10.

Part B · Proportional relationships (1 point each)

5. x: 4, 6, 10 → y: 10, 15, 25. Proportional? k?

6. x: 2, 4, 6 → y: 3, 7, 11. Proportional?

7. Write the equation: $8.25 per hour

8. y = 0.4x; x = 35

9. Is y = 7 + x proportional?

10. A proportional graph passes through (5, 40). k?

11. Interpret (1, 12) on a graph of hours vs miles.

Part C · Problem solving (3 points each)

12 A car travels 196 miles on 7 gallons. Write an equation for miles m and gallons g. How far can it go on a full 15-gallon tank?

13 Sunscreen: 3 oz for $7.20 or 8 oz for $17.60. Which is the better buy? By how much per ounce?

14 Lemonade is 2 parts juice to 7 parts water. How much juice is in 45 cups of lemonade?

UNIT 2

Percents

Percent as a rate per 100, percent equations, discounts, tax, tips, percent change, interest, and percent error.

Your practice pages for Unit 2. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

percent	A rate per 100.
discount	Amount off a price.
markup	Amount added to cost.
percent change	Change ÷ original.
simple interest	$I = P \times r \times t$.
percent error	|estimate − actual| ÷ actual.

UNIT 2 · LESSON 2.1 · PRACTICE

Lesson 2.1: What Is a Percent?

Name _____ Date _____

I can explain a percent as a rate per 100.

Warm-Up

1. 25 × 4

2. 18 × 4

Write as a percent

3. 13 out of 100

4. 9 out of 10

5. 12 out of 50

6. 3 out of 5

7. 7 out of 25

8. 1 out of 4

9. 20 out of 20

10. 11 out of 20

Solve

11. A basketball player made 17 of 20 free throws. What percent did she make?

12. Lillie answered 46 of 50 questions correctly. Her percent score?

Exit Ticket

13. Explain why 50% of something is half of it.

UNIT 2 · LESSON 2.2 · PRACTICE

Lesson 2.2: Percents, Fractions, and Decimals

Name _____ Date _____

I can write percents as fractions and decimals, and fractions and decimals as percents.

Warm-Up

1. $3 \div 8$

2. 0.4×100

Complete

3. 62% = ___ (decimal)

4. 8% = ___ (decimal)

5. 0.9 = ___%

6. 1.25 = ___%

7. $^{2}/_{5}$ = ___%

8. $^{5}/_{8}$ = ___%

9. 30% = ___ (fraction, simplest)

10. 64% = ___ (fraction)

Order from least to greatest

11. $^{2}/_{3}$, 0.65, 64%

Exit Ticket

12. Explain why 0.05 is 5%, not 50%.

UNIT 2 · LESSON 2.3 · PRACTICE

Lesson 2.3: Finding a Percent of a Number

Name _____ Date _____

I can find a percent of a number using models and multiplication.

Warm-Up

1. 10% of 60

2. 0.15 × 240

Find

3. 10% of 90

4. 25% of 64

5. 40% of 35

6. 5% of 180

7. 75% of 48

8. 12% of 250

9. 150% of 30

10. 1% of 600

Solve

11. A school has 450 students. 22% play an instrument. How many?

12. A $25 meal plus an 18% tip. How much is the tip?

Exit Ticket

13. Find 35% of 80 two different ways.

UNIT 2 · LESSON 2.4 · PRACTICE

Lesson 2.4: Finding What Percent

Name _____ Date _____

I can find what percent one number is of another.

Warm-Up

1. 42 ÷ 48

2. 0.875 = ___%

What percent?

3. 9 of 36

4. 24 of 30

5. 13 of 52

6. 45 of 60

7. 3 of 12

8. 66 of 55

Solve

9. A team won 27 of 36 games. What percent did it win?

10. A store sold 340 of its 400 jackets. What percent are left?

Exit Ticket

11. A shirt costs $12. A sweater costs $30. The shirt's price is what percent of the sweater's?

WEEKLY CHECK · Fri Dec 11 · ABOUT 15 MINUTES

Weekly Check 4: Percent Basics

Name _____ Date _____

Show what you know

1. 9 out of 20 = ___%

2. $^{3}/_{8}$ = ___%

3. 0.07 = ___%

4. 30% of 70

5. What percent of 60 is 45?

6. A team won 14 of 16 games. Percent won?

UNIT 2 · LESSON 2.5 · PRACTICE

Lesson 2.5: Finding the Whole

Name _____ Date _____

I can find the whole when I know a part and its percent.

Warm-Up

1. 45 ÷ 0.30

2. If 10% is 7, 100% is ___

Find the whole

3. 10% of ___ is 12

4. 25% of ___ is 9

5. 50% of ___ is 37

6. 60% of ___ is 42

7. 5% of ___ is 3

8. 125% of ___ is 50

Solve

9. A player scored 24 points, which was 40% of the team's points. How many points did the team score?

10. Ellie paid a $6 deposit, which is 15% of the price. What is the price?

Exit Ticket

11. 18 students walk to school. That is 12% of the school. How many students attend?

UNIT 2 · LESSON 2.6 · PRACTICE

Lesson 2.6: The Percent Equation

Name _____ Date _____

I can use part = percent × whole to solve any percent problem.

Warm-Up

1. 36 ÷ 0.45

2. 0.64 × 75

Write an equation and solve

3. What is 18% of 350?

4. 42 is what percent of 168?

5. 91 is 70% of what?

6. What is 2.5% of 400?

7. 60 is what percent of 48?

8. 15 is 6% of what?

Exit Ticket

9. Write and solve: 14 is 35% of what number?

UNIT 2 · LESSON 2.7 · PRACTICE

Lesson 2.7: Percents Greater Than 100% and Less Than 1%

Name _____ Date _____

I can work with percents greater than 100% and less than 1%.

Warm-Up

1. 2.5 × 80

2. 0.005 × 2,000

Find

3. 200% of 34

4. 110% of 60

5. 0.1% of 9,000

6. 0.75% of 400

7. 175% of 20

8. 0.4% as a decimal

Solve

9. A puppy's weight is now 300% of its birth weight of 2.5 lb. How much does it weigh?

10. A savings account earns 0.6% per year on $1,500. How much interest in 1 year?

Exit Ticket

11. Is 150% of a number more or less than the number? Explain.

UNIT 2 · LESSON 2.8 · PRACTICE

Lesson 2.8: Estimating with Percents

Name _____ Date _____

I can estimate percents using benchmarks like 10%, 25%, and 50%.

Warm-Up

1. 10% of 47.80

2. Half of 4.80

Estimate

3. 20% of $31.75

4. 25% of 397

5. 5% of $198

6. 33% of 61

7. Tip of 18% on $52

8. 75% of 81

Exit Ticket

9. Explain how to find 15% of a restaurant bill in your head.

WEEKLY CHECK · Fri Dec 18 · ABOUT 20 MINUTES

Weekly Check 5: Percent Equations

Name _____ Date _____

Show what you know

1. 40% of what is 26?

2. 18 is what percent of 24?

3. 250% of 18

4. 0.3% of 2,000

5. Estimate 15% of $79.

6. Write and solve: 63 is 35% of n.

UNIT 2 · LESSON 2.9 · PRACTICE

Lesson 2.9: Discounts and Sale Prices

Name _____ Date _____

I can find the sale price after a percent discount.

Warm-Up

1. 0.25 × 80

2. 100% − 15%

Find the sale price

3. 20% off $35.00

4. 30% off $50.00

5. 10% off $89.90

6. 35% off $60.00

7. 50% off $27.80

8. 12% off $250.00

Solve

9. A video game costs $59.99. It is 30% off. What is the sale price?

10. A $150 bike is on sale for $105. What is the percent discount?

Exit Ticket

11. Is 20% off and then another 20% off the same as 40% off? Try it with $100.

UNIT 2 · LESSON 2.10 · PRACTICE

Lesson 2.10: Markups and Sales Tax

Name _____ Date _____

I can find prices after a markup or sales tax.

Warm-Up

1. 0.07 × 40

2. 1.6 × 25

Find the total

3. 7% tax on $20.00

4. 7% tax on $64.00

5. 5% tax on $38.00

6. 50% markup on $18.00

7. 25% markup on $64.00

8. 120% markup on $15.00

Solve

9. A bookstore buys a book for $12 and sells it for $18. What is the percent markup?

10. Lillie buys $28.50 of supplies with 7% tax. What does she pay?

Exit Ticket

11. A $50 item has a 20% markup, then a 20% discount. What is the final price?

UNIT 2 · LESSON 2.11 · PRACTICE

Lesson 2.11: Percent Increase and Decrease

Name _____ Date _____

I can find the percent of increase or decrease.

Warm-Up

1. 6 ÷ 8

2. 300 ÷ 2,500

Find the percent change

3. 20 → 25

4. 50 → 35

5. 12 → 18

6. 200 → 170

7. 64 → 80

8. 90 → 63

Solve

9. A gallon of milk went from $3.20 to $3.60. Percent increase?

10. A plant was 15 cm tall and is now 24 cm. Percent increase?

Exit Ticket

11. A price goes from $50 to $100, then back to $50. Are the percent changes the same? Explain.

UNIT 2 · LESSON 2.12 · PRACTICE

Lesson 2.12: Tips, Commissions, and Fees

Name _____ Date _____

I can solve problems with tips, commissions, and fees.

Warm-Up

1. 0.06 × 1,250

2. 0.18 × 42

Solve

3. 20% tip on a $55 dinner. Total?

4. 5% commission on a $24,000 car

5. A $2 fee plus 10% on a $30 ticket. Total?

6. Lillie earns $8 per hour plus 10% of her sales. She works 5 hours and sells $300. Pay?

7. 15% tip on a $28.40 haircut

8. A real estate agent earns 3% on a $180,000 house

Exit Ticket

9. Which earns more: 8% commission on $2,500 or a flat $175?

WEEKLY CHECK · Fri Jan 8 · ABOUT 20 MINUTES

Weekly Check 6: Discounts, Tax, and Change

Name _____ Date _____

Show what you know

1. 35% off $80

2. 7% tax on $45

3. $30 → $36 percent change

4. $250 → $200 percent change

5. 18% tip on $50

6. 6% commission on $3,500

UNIT 2 · LESSON 2.13 · PRACTICE

Lesson 2.13: Simple Interest

Name _____ Date _____

I can find simple interest and the total amount in an account or loan.

Warm-Up

1. 500 × 0.03

2. 15 × 4

Find the interest and total

3. $400.00 at 5% for 2 years

4. $1,500.00 at 3% for 4 years

5. $250.00 at 8% for 1 years

6. $6,000.00 at 2.5% for 3 years

Solve

7. Which earns more: $1,000 at 4% for 3 years or $1,200 at 3% for 3 years?

Exit Ticket

8. Explain why borrowing money costs more than the amount borrowed.

UNIT 2 · LESSON 2.14 · PRACTICE

Lesson 2.14: Percent Error

Name _____ Date _____

I can find the percent error of an estimate or measurement.

Warm-Up

1. |120 − 150|

2. 30 ÷ 150

Find the percent error

3. Estimate 90, actual 100

4. Estimate 55, actual 50

5. Estimate 18, actual 24

6. Estimate 420, actual 400

7. Estimate 3.6, actual 4

8. Estimate 130, actual 125

Exit Ticket

9. Lillie estimated a room was 12 ft long. It was 15 ft. Find the percent error.

UNIT 2 · LESSON 2.15 · PRACTICE

Lesson 2.15: Multi-Step Percent Problems

Name _____ Date _____

I can solve problems that combine discounts, tax, tips, and percent change.

Warm-Up

1. 0.75 × 60

2. 1.07 × 45

Solve

3. A $120 jacket is 30% off. Tax is 7%. Total?

4. A family's dinner is $64. Tax is 7% and they tip 18% of $64. Total?

5. A stock worth $50 rises 20% and then falls 15%. What is it worth?

6. A store marks up a $30 item 50%, then puts it on sale for 20% off. Sale price?

Exit Ticket

7. Does the order of a 10% discount and 7% tax matter for the final price? Test with $100.

UNIT 2 · LESSON 2.16 · PRACTICE

Lesson 2.16: Percent Problem Solving

Name _____ Date _____

I can choose and use the right percent strategy to solve real problems.

Warm-Up

1. 0.8 × 35

2. 1.07 × 64

Mixed practice

3. 35% of 260

4. 54 is what percent of 72?

5. 27 is 45% of what?

6. Sale price: 15% off $86

7. I for $900 at 4% for 5 years

8. Percent error: estimate 38, actual 40

Exit Ticket

9. Write a multi-step percent problem about something you want to buy. Solve it.

UNIT REVIEW · Thu Jan 14 (homework) or as needed · ABOUT 40 MINUTES

Unit 2 Review

Name _____ Date _____

Percent basics

1. 0.45 = ___%

2. $^{7}/_{8}$ = ___%

3. 24% of 150

4. What percent of 80 is 68?

5. 15 is 60% of what?

Applications

6. 25% off $64, then 7% tax

7. Percent change: 40 → 52

8. I = ? for $2,500 at 3% for 4 years

9. Percent error: guess 90, actual 75

UNIT TEST · Fri Jan 15 · ABOUT 45 MINUTES

Unit 2 Test

Name _____ Date _____

Show your work. This test has 23 points.

Part A · Percent basics (1 point each)

1. 13 out of 25 = ___%

2. 1.8 = ___%

3. $^{5}/_{8}$ = ___%

4. 0.4% as a decimal

Part B · Percent equations (1 point each)

5. 16% of 75

6. What percent of 120 is 42?

7. 51 is 85% of what?

8. 150% of 46

9. Estimate 18% of $41

Part C · Applications (2 points each)

10. A $72 jacket is 35% off. Sale price?

11. A $36 meal with 7% tax and a 20% tip on $36. Total?

12. A town grew from 4,800 to 5,520 people. Percent increase?

13. $1,800 at 3.5% simple interest for 3 years. Interest?

Part D · Problem solving (3 points each)

14 Store A sells a $90 bike at 30% off. Store B sells the same bike for $70 with a $5 coupon. Which is cheaper, and by how much?

15 Lillie guessed a pumpkin weighed 12 lb. It weighed 15 lb. Ellie guessed 17 lb. Who had the smaller percent error?

UNIT 3

Adding and Subtracting Integers

Opposites, absolute value, adding and subtracting integers with chips and number lines, and real-world applications.

Your practice pages for Unit 3. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

integer	Whole numbers and their opposites.
opposites	Same distance from 0, opposite sides.
absolute value	Distance from 0.
additive inverse	A number's opposite; sums to 0.

UNIT 3 · LESSON 3.1 · PRACTICE

Lesson 3.1: Integers, Opposites, and Absolute Value

Name _____ Date _____

I can use integers to describe real situations and find opposites and absolute values.

Warm-Up

1. Which is colder: −3°F or −8°F?

2. Name a number between −2 and 0.

Write an integer

3. a deposit of $40

4. 20 feet below sea level

5. a temperature drop of 6 degrees

6. a gain of 3 yards

Find

7. opposite of 31

8. opposite of −4

9. $|-18|$

10. $|0|$

11. $-|-6|$

12. $|14| - |-5|$

Order from least to greatest

13. 3, −7, 0, −2, 5

Exit Ticket

14 Can absolute value be negative? Explain.

UNIT 3 · LESSON 3.2 · PRACTICE

Lesson 3.2: Adding Integers with the Same Sign

Name _____ Date _____

I can add two integers that have the same sign.

Warm-Up

1. $|-9| + |-6|$

2. $4 + 3$

Add

3. $-3 + (-4)$

4. $-11 + (-2)$

5. $6 + 8$

6. $-7 + (-7)$

7. $-15 + (-9)$

8. $-24 + (-36)$

9. $13 + 27$

10. $-50 + (-50)$

Solve

11. A diver is at −12 m and dives 8 m deeper. Where is she now?

12. The temperature is −5°F and drops 9 more degrees. New temperature?

Exit Ticket

13. Use chips to show $-4 + (-5)$. What is the sum?

UNIT 3 · LESSON 3.3 · PRACTICE

Lesson 3.3: Adding Integers with Different Signs

Name _____ Date _____

I can add two integers with different signs.

Warm-Up

1. $-4 + (-5)$

2. $14 - 9$

Add

3. $5 + (-2)$

4. $-5 + 2$

5. $9 + (-13)$

6. $-12 + 20$

7. $30 + (-45)$

8. $-17 + 17$

9. $-8 + 3$

10. $21 + (-6)$

Solve

11. Your account has $35. You buy a $52 game with an overdraft. Balance?

12. A football team gains 6 yards, then loses 9. Total?

Exit Ticket

13. Explain why $-7 + 7 = 0$ using zero pairs.

UNIT 3 · LESSON 3.4 · PRACTICE

Lesson 3.4: Additive Inverses

Name _____ Date _____

I can explain why a number plus its opposite equals 0.

Warm-Up

1. $-6 + 6$

2. $15 + (-15)$

Find the sum using inverses

3. $-7 + 12 + 7$

4. $25 + (-18) + (-25)$

5. $-3 + 9 + (-9) + 3$

6. $40 + (-11) + (-40)$

Missing number

7. $-8 + ___ = 0$

8. $___ + (-26) = 0$

9. $-5 + ___ = 2$

10. $14 + ___ = -1$

Exit Ticket

11. Describe a real situation where two opposite quantities combine to make 0.

WEEKLY CHECK · Fri Jan 22 · ABOUT 15 MINUTES

Weekly Check 7: Adding Integers

Name _____ Date _____

Show what you know

1. $-9 + (-14)$

2. $-9 + 14$

3. $21 + (-30)$

4. $|-17|$

5. $-13 + ___ = 0$

6. $-6 + 11 + 6$

UNIT 3 · LESSON 3.5 · PRACTICE

Lesson 3.5: Adding Integers on a Number Line

Name _____ Date _____

I can use a number line to add integers.

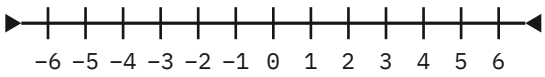
Warm-Up

1. $-7 + 4$

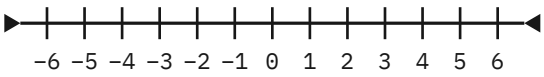
2. $-2 + (-6)$

Use the number line

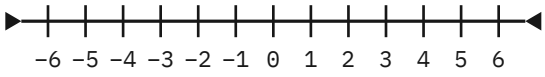
3. $-5 + 8$



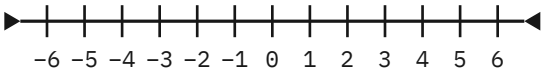
4. $4 + (-9)$



5. $-1 + (-4) + 3$



6. $6 + (-2) + (-7)$



Exit Ticket

7. Draw a number line for $-6 + 10$ and give the sum.

UNIT 3 · LESSON 3.6 · PRACTICE

Lesson 3.6: Subtracting Integers

Name _____ Date _____

I can subtract integers by adding the opposite.

Warm-Up

1. Opposite of −9

2. −6 + (−4)

Subtract

3. 7 − 10

4. −3 − 5

5. 4 − (−6)

6. −9 − (−2)

7. −1 − (−8)

8. 0 − (−15)

9. 12 − (−12)

10. −20 − 13

Rewrite as addition, then solve

11. 15 − (−7)

12. −11 − 4

Exit Ticket

13. Explain why 6 − (−2) is greater than 6.

UNIT 3 · LESSON 3.7 · PRACTICE

Lesson 3.7: Subtraction as Distance

Name _____ Date _____

I can use a number line to show subtraction as the change from one number to another.

Warm-Up

1. $11 - (-4)$

2. $-8 - 25$

Find the change (end − start)

3. from −2 to 6

4. from 8 to −1

5. from −15 to −5

6. from 0 to −9

7. from −7 to −20

8. from 14 to −3

Solve

9. A submarine at −120 m rises to −45 m. Change?

10. A company's profit goes from $3 million to −$2 million. What is the change?

Exit Ticket

11. Explain how a number line shows $4 - (-3) = 7$.

UNIT 3 · LESSON 3.8 · PRACTICE

Lesson 3.8: Distance Between Two Numbers

Name _____ Date _____

I can find the distance between two numbers using absolute value.

Warm-Up

1. $|-8|$

2. $3 - (-5)$

Find the distance

3. -3 and 4

4. 10 and -10

5. -25 and -9

6. 0 and -14

7. -6 and 6

8. -100 and 35

Solve

9. Highest Mississippi point: Woodall Mountain, 807 ft. Lowest: sea level on the Gulf, 0 ft. Difference?

10. Low of $-11°F$ and high of $23°F$. Temperature range?

Exit Ticket

11. Why do we use absolute value for distance?

WEEKLY CHECK · Fri Jan 29 · ABOUT 15 MINUTES

Weekly Check 8: Subtracting Integers

Name _____ Date _____

Show what you know

1. $4 - 11$

2. $-7 - (-3)$

3. $-5 - 8$

4. $10 - (-6)$

5. Distance between -8 and 5

6. From $-6°F$ to $9°F$: change?

UNIT 3 · LESSON 3.9 · PRACTICE

Lesson 3.9: Integers: Temperature and Elevation

Name _____ Date _____

I can solve temperature and elevation problems with integers.

Warm-Up

1. $-6 + 23$

2. $9 - 5$

Solve

3. The temperature was −12°F and rose 19°. New temperature?

4. A plane at 2,500 ft descends 3,100 ft over the Dead Sea basin. New elevation?

5. A miner is 450 ft underground and goes up 120 ft. Elevation?

6. Morning −5°F, afternoon 14°F. How much warmer?

Exit Ticket

7. Write a temperature story for $-8 + 13$ and solve it.

UNIT 3 · LESSON 3.10 · PRACTICE

Lesson 3.10: Integers and Money

Name _____ Date _____

I can use integers to track money: deposits, withdrawals, profits, and debts.

Warm-Up

1. 25 + (−40)

2. −15 + 30

Track the balance

3. Start $0. Earn $45, spend $28, spend $30, earn $20.

4. Start −$12. Deposit $50, withdraw $65.

Solve

5. A business lost $1,200 in January and earned $2,050 in February. Total?

6. You owe $35 and pay back $22. How much do you owe now?

Exit Ticket

7. Make a 4-transaction account record that ends at −$5.

UNIT 3 · LESSON 3.11 · PRACTICE

Lesson 3.11: Multi-Step Integer Problems

Name _____ Date _____

I can solve multi-step problems with integer addition and subtraction.

Warm-Up

1. 12 + 25

2. −7 + (−3)

Evaluate

3. −6 + 9 − 4

4. 15 − 22 + 3

5. −10 − 5 + 18

6. 7 − (−3) − 12

7. −4 + (−8) − (−6)

8. 20 − 35 + 10 − 5

Solve

9. A hiker starts at −40 ft (in a valley), climbs 125 ft, descends 60 ft, climbs 15 ft. Elevation?

Exit Ticket

10. Write a 3-step temperature story and solve it.

UNIT 3 · LESSON 3.12 · PRACTICE

Lesson 3.12: Properties and Mental Math with Integers

Name _____ Date _____

I can use properties of addition to add integers mentally.

Warm-Up

1. $-17 + 17$

2. $45 + (-5)$

Use properties to evaluate mentally

3. $-8 + 27 + 8$

4. $45 + (-19) + (-45)$

5. $-13 + (-7) + 50$

6. $64 - 30 - 34$

7. $-99 + 50 + 99 + (-50)$

8. $125 + (-75) + (-25)$

Exit Ticket

9. Show two ways to group $-6 + 16 + (-4)$.

WEEKLY CHECK · Fri Feb 5 · ABOUT 15 MINUTES

Weekly Check 9: Integer Applications

Name _____ Date _____

Show what you know

1. −15 m diver rises 9 m

2. $20 balance, spend $32

3. −4 + 10 − 9 + 2

4. −29 + 46 + 29 (mentally)

5. Temp 6°F, drops 11°, rises 3°

6. Profits $40, −$55, $30: total

UNIT 3 · LESSON 3.13 · PRACTICE

Lesson 3.13: Integer Expressions

Name _____ Date _____

I can evaluate expressions with integers and variables.

Warm-Up

1. $-6 - (-11)$

2. $-(-4)$

Evaluate when $x = -5$ and $y = 3$

3. $x + y$

4. $x - y$

5. $y - x$

6. $-x$

7. $x + x + y$

8. $|x| - y$

Exit Ticket

9. Find a value of x so that $x - 8$ is negative and $x + 8$ is positive.

UNIT 3 · LESSON 3.14 · PRACTICE

Lesson 3.14: Explaining Integer Rules with Models

Name _____ Date _____

I can explain integer addition and subtraction rules using chips and number lines.

Warm-Up

1. $4 - (-2)$

2. $-3 + 5$

Explain with a model and solve

3. $-5 + 2$

4. $3 - 8$

5. $-4 - (-6)$

6. $0 - (-3)$

Exit Ticket

7. Which model do you prefer and why?

UNIT 3 · LESSON 3.15 · PRACTICE

Lesson 3.15: Finding and Fixing Integer Errors

Name _____ Date _____

I can find and fix mistakes in integer calculations.

Warm-Up

1. $-8 - 3$

2. $5 - (-2)$

Find and fix the error

3. $-12 + 5 = -17$

4. $9 - (-4) = 5$

5. $-3 - 3 = 0$

6. $-10 + 15 = -5$

7. $0 - 8 = 8$

8. $|-2| + |-3| = -5$

Exit Ticket

9. What is the most common integer mistake you make? How will you avoid it?

UNIT 3 · LESSON 3.16 · PRACTICE

Lesson 3.16: Integer Problem Solving

Name _____ Date _____

I can solve problems with integer addition and subtraction from start to finish.

Warm-Up

1. $3 - (-7)$

2. $5 + (-17)$

Mixed practice

3. $-23 + 15$

4. $-23 - 15$

5. $15 - (-23)$

6. $|-9 - 4|$

7. $-6 + 13 - (-5) - 20$

8. $x - y$ when $x = -2$, $y = -9$

Solve

9. A submarine at −250 ft rises 90 ft, dives 145 ft, then rises 60 ft. Final depth?

Exit Ticket

10. Write and solve a problem that uses both addition and subtraction of negatives.

UNIT REVIEW · Thu Feb 11 (homework) or as needed · ABOUT 35 MINUTES

Unit 3 Review

Name _____ Date _____

Add and subtract

1. −18 + (−7)
2. 25 + (−40)
3. −12 − 9
4. 6 − (−14)
5. −3 − (−10)
6. |−5 − 12|

Applications

7. −8°F rises 21°
8. Elevation −45 ft to 30 ft: change
9. x − y + 3 when x = −1, y = −6
10. Fix: −9 − (−4) = −13

UNIT TEST · Fri Feb 12 · ABOUT 40 MINUTES

Unit 3 Test

Name _____ Date _____

Show your work. This test has 21 points.

Part A · Integers (1 point each)

1. Opposite of −21

2. |−34|

3. Order: 4, −9, 0, −3

4. −17 + ___ = 0

Part B · Compute (1 point each)

5. −15 + (−26)

6. −38 + 22

7. 44 + (−19)

8. 9 − 25

9. −14 − (−30)

10. −6 − 18

11. |−11 − 7|

12. −8 + 20 − (−5) − 13

Part C · Applications (2 points each)

13. At 5 a.m. it was −9°F. By 2 p.m. it was 16°F. How much did it rise?

14. A diver at −42 ft rises 15 ft, then dives 28 ft. Where is she?

15. Evaluate a − b + c for a = −3, b = −10, c = −8.

Part D · Explain (3 points)

16 Use a chip model or number line to explain why 2 − (−6) = 8.

UNIT 4

Adding and Subtracting Rational Numbers

4

Signed fractions and decimals, converting fractions to decimals, and real-world problems.

Your practice pages for Unit 4. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

rational number	A number that can be written as a fraction of integers.
terminating decimal	A decimal that ends.
repeating decimal	A decimal with a repeating pattern.

UNIT 4 · LESSON 4.1 · PRACTICE

Lesson 4.1: Rational Numbers on the Number Line

Name _____ Date _____

I can place positive and negative fractions and decimals on a number line.

Warm-Up

1. Which is less: −5 or −2?

2. $\frac{3}{4}$ as a decimal

Compare (write <, >, or =)

3. $-0.6 \bigcirc -0.06$

4. $-\frac{1}{2} \bigcirc -0.5$

5. $-\frac{2}{5} \bigcirc -\frac{1}{3}$

6. $-3.4 \bigcirc -3.45$

7. $\frac{5}{8} \bigcirc -\frac{7}{8}$

8. $-1.75 \bigcirc -1\frac{3}{4}$

Order from least to greatest

9. $0.5, -\frac{3}{4}, -0.7, \frac{1}{4}, -1$

Exit Ticket

10. Explain why −2.1 is less than −2.

UNIT 4 · LESSON 4.2 · PRACTICE

Lesson 4.2: Adding Signed Fractions

Name _____ Date _____

I can add positive and negative fractions.

Warm-Up

1. $-5 + 3$

2. $\frac{1}{2} + \frac{1}{3}$

Add

3. $-\frac{3}{7} + \frac{1}{7}$

4. $\frac{5}{9} + (-\frac{8}{9})$

5. $-\frac{1}{4} + (-\frac{3}{8})$

6. $\frac{2}{3} + (-\frac{1}{6})$

7. $-\frac{7}{10} + \frac{2}{5}$

8. $-\frac{5}{12} + (-\frac{1}{3})$

9. $\frac{4}{5} + (-\frac{1}{2})$

10. $-\frac{3}{4} + \frac{3}{4}$

Solve

11. A water level was $-\frac{3}{4}$ ft (below normal). It rose $\frac{1}{2}$ ft. New level?

Exit Ticket

12. $-\frac{5}{6} + \frac{3}{4}$

UNIT 4 · LESSON 4.3 · PRACTICE

Lesson 4.3: Subtracting Signed Fractions

Name _____ Date _____

I can subtract positive and negative fractions by adding the opposite.

Warm-Up

1. $2 - (-5)$

2. Opposite of $-\frac{3}{8}$

Subtract

3. $\frac{1}{5} - \frac{4}{5}$

4. $-\frac{2}{9} - \frac{5}{9}$

5. $\frac{3}{8} - (-\frac{1}{4})$

6. $-\frac{5}{6} - (-\frac{1}{3})$

7. $-\frac{1}{2} - \frac{1}{4}$

8. $\frac{2}{3} - \frac{3}{4}$

9. $-\frac{7}{12} - (-\frac{3}{4})$

10. $0 - (-\frac{5}{8})$

Exit Ticket

11. Explain why $\frac{1}{2} - (-\frac{1}{2}) = 1$.

UNIT 4 · LESSON 4.4 · PRACTICE

Lesson 4.4: Adding and Subtracting Signed Mixed Numbers

Name _____ Date _____

I can add and subtract signed mixed numbers like $-2\frac{1}{3} + 1\frac{1}{2}$.

Warm-Up

1. $2\frac{1}{3}$ as an improper fraction

2. $\frac{17}{6}$ as a mixed number

Compute

3. $-2\frac{1}{2} + 4\frac{1}{4}$

4. $3\frac{1}{3} + (-3\frac{5}{6})$

5. $-1\frac{5}{8} + (-\frac{3}{4})$

6. $-3\frac{1}{2} - 1\frac{1}{4}$

7. $2\frac{1}{4} - (-3\frac{2}{3})$

8. $-3\frac{1}{5} - (-2\frac{1}{10})$

Solve

9. A lake was $1\frac{1}{4}$ ft below normal. It dropped $2\frac{2}{3}$ ft more. New level?

Exit Ticket

10. $-2\frac{1}{2} - (-5\frac{3}{4})$

WEEKLY CHECK · Fri Feb 19 · ABOUT 20 MINUTES

Weekly Check 10: Signed Fractions

Name _____ Date _____

Show what you know

1. $-\frac{3}{5} + \frac{1}{5}$

2. $-\frac{1}{4} + \left(-\frac{2}{3}\right)$

3. $\frac{1}{6} - \frac{5}{6}$

4. $-\frac{3}{8} - \left(-\frac{1}{2}\right)$

5. $-3\frac{1}{2} + 2\frac{1}{4}$

6. Order: $-\frac{2}{3}$, -0.6, $\frac{1}{2}$

UNIT 4 · LESSON 4.5 · PRACTICE

Lesson 4.5: Fractions as Decimals

Name _____ Date _____

I can write fractions as terminating or repeating decimals using long division.

Warm-Up

1. $3 \div 8$

2. $1 \div 3$

Write as a decimal. Terminating or repeating?

3. $\frac{3}{5}$

4. $\frac{1}{6}$

5. $\frac{7}{8}$

6. $\frac{2}{9}$

7. $\frac{9}{20}$

8. $\frac{4}{11}$

Exit Ticket

9. Write $-2\frac{1}{3}$ as a decimal.

UNIT 4 · LESSON 4.6 · PRACTICE

Lesson 4.6: Comparing and Ordering Rational Numbers

Name _____ Date _____

I can compare and order signed fractions and decimals.

Warm-Up

1. $5/8$ as a decimal

2. $4/3$ as a decimal

Order from least to greatest

3. $-0.4, -3/8, 0.3, -1/2$

4. $1\,1/3, -1.3, 1.3, -1\,1/3$

5. $-2.05, -2.5, -2\,1/5, -2$

Exit Ticket

6. Which is greater, $-2/3$ or -0.67? Explain.

UNIT 4 · LESSON 4.7 · PRACTICE

Lesson 4.7: Adding Signed Decimals

Name _____ Date _____

I can add positive and negative decimals.

Warm-Up

1. 40.75 − 12.50

2. 3.6 + 0.85

Add

3. −4.3 + 2.1

4. 6.75 + (−9.5)

5. −0.8 + (−0.15)

6. −15.2 + 15.2

7. 3.05 + (−0.7)

8. −22.4 + (−7.65)

Solve

9. A temperature of −3.8°C rises 5.6°C. New temperature?

10. A stock changed −1.35, +0.80, and −0.25 dollars. Total change?

Exit Ticket

11. −8.06 + 5.9

UNIT 4 · LESSON 4.8 · PRACTICE

Lesson 4.8: Subtracting Signed Decimals

Name _____ Date _____

I can subtract positive and negative decimals.

Warm-Up

1. 163.1 + 2.4

2. −0.75 + (−1.6)

Subtract

3. 2.5 − 7.1

4. −6.3 − 2.4

5. 1.75 − (−0.5)

6. −8.2 − (−8.2)

7. −0.05 − 0.5

8. 10 − (−3.75)

Solve

9. From −12.5°F to 4.3°F: change?

10. A submarine at −85.4 m dives to −112.9 m. Change?

Exit Ticket

11. −5.08 − (−7.3)

WEEKLY CHECK · Fri Feb 26 · ABOUT 20 MINUTES

Weekly Check 11: Signed Decimals

Name _____ Date _____

Show what you know

1. $\frac{7}{8}$ as a decimal

2. $\frac{5}{9}$ as a decimal

3. $-6.4 + 2.75$

4. $-0.35 + (-1.2)$

5. $3.1 - 5.85$

6. $-2.5 - (-7.25)$

UNIT 4 · LESSON 4.9 · PRACTICE

Lesson 4.9: Rational Numbers and Money

Name _____ Date _____

I can solve money problems with positive and negative decimals.

Warm-Up

1. 24.6 − 18.95

2. 5.65 − 7.4

Solve

3. Start at −$8.25. Deposit $30, spend $14.60.

4. A lemonade stand: +$18.50, −$6.75 (supplies), +$12.25, −$4.30. Total?

5. A stock is at $45.20. It changes −2.85, +1.10, −0.65. New price?

6. Ellie owes $12.40 and pays $7.95. How much does she owe now?

Exit Ticket

7. Create a 4-transaction account record that ends below $0.

UNIT 4 · LESSON 4.10 · PRACTICE

Lesson 4.10: Rational Numbers in Measurement

Name _____ Date _____

I can solve elevation, temperature, and measurement problems with signed fractions and decimals.

Warm-Up

1. $5\frac{1}{4} - 4\frac{3}{8}$

2. $7.2 - (-4.5)$

Solve

3. A diver at $-10\frac{1}{2}$ ft rises $4\frac{3}{4}$ ft. Depth?

4. A runner's times: 12.45 s and 11.9 s. Improvement?

5. −6.25°F drops 3.8°F. New temperature?

6. Rope $12\frac{1}{3}$ m; cut $4\frac{5}{6}$ m. Left?

Exit Ticket

7. Write a measurement story for $2\frac{1}{2} - 3\frac{3}{4}$. Solve.

UNIT 4 · LESSON 4.11 · PRACTICE

Lesson 4.11: Properties for Rational Number Sums

Name _____ Date _____

I can use properties to add and subtract rational numbers efficiently.

Warm-Up

1. $-\frac{3}{4} + \frac{3}{4}$

2. $2.6 - 0.6$

Compute mentally

3. $4.8 + (-2.3) + (-4.8)$

4. $-\frac{5}{6} + 3 + \frac{5}{6}$

5. $-1.25 + 7 + (-0.75)$

6. $2\frac{1}{2} - 4 + \frac{1}{2}$

7. $-9.9 + 5 + 9.9 + (-5)$

8. $0.75 + (-\frac{1}{4}) + (-2)$

Exit Ticket

9. Show how grouping helps evaluate $-6.4 + 3.3 + 6.4 - 1.3$.

UNIT 4 · LESSON 4.12 · PRACTICE

Lesson 4.12: Multi-Step Rational Number Problems

Name _____ Date _____

I can solve multi-step problems with signed fractions and decimals.

Warm-Up

1. $2 - 5 + 10 - 3$

2. $\frac{4}{4}$

Solve

3. Temperatures: −2.4°C, then +6.8°C, then −5.1°C. Final?

4. A stock: start $18.40. Changes: $-\frac{3}{4}$, +$1.20, −$0.55. Final price? (Use 0.75 for $\frac{3}{4}$.)

5. Lillie's garden bed is $8\frac{1}{2}$ ft. She uses $2\frac{3}{4}$ ft for tomatoes and $3\frac{1}{3}$ ft for peppers. Space left?

Exit Ticket

6. Write and solve a three-step problem with negative decimals.

WEEKLY CHECK · Fri Mar 5 · ABOUT 20 MINUTES

Weekly Check 12: Rational Number Applications

Name _____ Date _____

Show what you know

1. Start −$6.40, deposit $25, spend $11.85

2. −3.2°C to 5.9°C: change

3. Board $6\frac{1}{4}$ ft, cut $2\frac{7}{8}$ ft. Left?

4. −4.5 + 2.75 + 4.5 (mentally)

5. −1.6 + 3.25 − 0.9

6. $\frac{1}{2} - \frac{3}{4} - \frac{1}{8}$

UNIT 4 · LESSON 4.13 · PRACTICE

Lesson 4.13: Distance Between Rational Numbers

Name _____ Date _____

I can find the distance between two rational numbers using absolute value.

Warm-Up

1. $|-7.5|$

2. $1\frac{3}{4} + 2\frac{1}{2}$

Find the distance

3. -4.6 and 2.9

4. $-\frac{2}{3}$ and $\frac{1}{6}$

5. -10.25 and -3.5

6. $3\frac{1}{2}$ and $-1\frac{1}{4}$

Solve

7. Death Valley: -86 m. Mount McKinley (Denali): 6,190 m. Distance?

Exit Ticket

8. Explain why the distance from -2.5 to 4 equals the distance from 4 to -2.5.

UNIT 4 · LESSON 4.14 · PRACTICE

Lesson 4.14: Expressions with Rational Numbers

Name _____ Date _____

I can evaluate expressions with signed fractions and decimals.

Warm-Up

1. $-1.75 + 3$

2. $-(-4.2)$

Evaluate when $p = -2.5$ and $q = 1.25$

3. $p + q$

4. $p - q$

5. $q - p$

6. $-p - q$

7. $p + p + q$

8. $|p| - |q|$

Exit Ticket

9. Evaluate $r - s$ when $r = {}^{2}/_{3}$ and $s = -{}^{5}/_{6}$.

UNIT 4 · LESSON 4.15 · PRACTICE

Lesson 4.15: Error Analysis with Rational Numbers

Name _____ Date _____

I can find and fix errors in rational-number computations.

Warm-Up

1. $-\frac{1}{2} - \frac{1}{3}$

2. $-2.5 + 1.3$

Find and fix

3. $-6.1 + 2.4 = -8.5$

4. $\frac{2}{3} - \frac{5}{6} = \frac{1}{6}$

5. $-4 - (-1.25) = -5.25$

6. $0.3 - 0.45 = 0.15$

Exit Ticket

7. Describe how estimating first can catch a sign error.

UNIT 4 · LESSON 4.16 · PRACTICE

Lesson 4.16: Rational Number Problem Solving

Name _____ Date _____

I can solve problems with signed fractions and decimals.

Warm-Up

1. $-4 - 1.5$

2. $-7.75 + 0.75$

Mixed practice

3. $-\frac{7}{8} + \frac{1}{4}$

4. $\frac{2}{5} - \left(-\frac{1}{2}\right)$

5. $-12.4 + 8.75$

6. $0.6 - 2.15$

7. Order: -0.3, $-\frac{1}{3}$, -0.33

8. $\frac{7}{12}$ as a decimal

Solve

9. A business had monthly results of −$1,250.50, +$3,400.75, and −$875.25. Total?

Exit Ticket

10. Which is harder for you: signed fractions or signed decimals? Make up and solve one of each.

UNIT REVIEW · Thu Mar 11 (homework) or as needed · ABOUT 35 MINUTES

Unit 4 Review

Name _____ Date _____

Compute

1. $-\frac{5}{6} + \frac{1}{3}$

2. $-\frac{3}{4} - \frac{5}{8}$

3. $-2\frac{3}{4} + (-1\frac{1}{2})$

4. $-9.6 + 4.35$

5. $-1.2 - (-3.05)$

6. $\frac{3}{11}$ as a decimal

Apply

7. Order: -1.5, $-1\frac{2}{3}$, -1.6

8. Distance between -2.75 and 4.1

9. $x - y$ when $x = -0.4$, $y = 1.35$

10. A balance of $-\$15.60$ gets a $\$40$ deposit and a $\$9.25$ charge.

UNIT TEST · Fri Mar 12 · ABOUT 45 MINUTES

Unit 4 Test

Name _____ Date _____

Show your work. This test has 21 points.

Part A · Rational numbers (1 point each)

1. $\frac{9}{16}$ as a decimal

2. $\frac{2}{3}$ as a decimal

3. Order: -0.8, $-\frac{3}{4}$, $-\frac{5}{6}$

4. $|-6.05|$

Part B · Compute (1 point each)

5. $-\frac{4}{9} + \left(-\frac{2}{3}\right)$

6. $\frac{7}{10} + \left(-1\frac{1}{5}\right)$

7. $\frac{3}{8} - \left(-\frac{5}{12}\right)$

8. $-4\frac{1}{4} - 1\frac{5}{6}$

9. $-14.25 + 6.8$

10. $0.45 - 2.6$

11. $-8.3 - (-12.05)$

12. $-3.75 + 9.2 + 3.75$ (mentally)

Part C · Applications (2 points each)

13. Temperatures change $-2.6°$, $+4.15°$, $-0.8°$ from $-1.5°$C. Final temperature?

14. A diver at $-12\frac{1}{2}$ ft rises $7\frac{3}{4}$ ft. Depth now?

15. Distance between -18.4 m and 22.75 m

Part D · Explain (3 points)

16 Ellie says $-2\frac{1}{4} + 1\frac{3}{4} = -3\frac{2}{4}$. Explain her mistake and give the correct answer.

UNIT 5

Multiplying and Dividing Rational Numbers

Sign rules and why they work, signed fractions and decimals, order of operations, rates, and averages.

Your practice pages for Unit 5. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

product	Result of multiplication.
quotient	Result of division.
reciprocal	Number whose product with another is 1.
mean	Sum ÷ count.

UNIT 5 · LESSON 5.1 · PRACTICE

Lesson 5.1: Multiplying Integers: Patterns

Name _____ Date _____

I can use patterns to discover how to multiply positive and negative integers.

Warm-Up

1. $-3 + (-3) + (-3) + (-3)$

2. 6×7

Multiply

3. $3 \times (-8)$

4. $(-5) \times 7$

5. $(-6) \times (-6)$

6. $(-11) \times 2$

7. $(-4) \times (-9)$

8. $12 \times (-3)$

9. $(-1) \times (-15)$

10. $0 \times (-23)$

Complete the pattern

11. $3 \times (-5) = -15$, $2 \times (-5) =$ ___, $1 \times (-5) =$ ___, $0 \times (-5) =$ ___, $-1 \times (-5) =$ ___

Exit Ticket

12. Use a pattern to explain why $-3 \times (-4)$ is positive.

UNIT 5 · LESSON 5.2 · PRACTICE

Lesson 5.2: Multiplying Integers: Rules and Reasons

Name _____ Date _____

I can multiply integers, including more than two factors, and explain the sign rules.

Warm-Up

1. $(-4) \times 5$

2. $(-20) \times (-2)$

Multiply

3. $(-2)(6)(-3)$

4. $(-5)(-2)(-4)$

5. $(7)(-1)(3)$

6. $(-3)(-3)(-3)$

7. $(-10)(-10)$

8. $(-2)(-2)(-2)(-2)$

Sign only

9. Is the product of 9 negative numbers positive or negative?

10. Is $(-8)(-7)(-6)(-5)$ positive or negative?

Exit Ticket

11. Explain why $(-4)^2$ and -4^2 are different.

UNIT 5 · LESSON 5.3 · PRACTICE

Lesson 5.3: Dividing Integers

Name _____ Date _____

I can divide integers using the related multiplication fact.

Warm-Up

1. $5 \times (-12)$

2. $(-5) \times 12$

Divide

3. $-36 \div 4$

4. $45 \div (-9)$

5. $-72 \div (-8)$

6. $100 \div (-25)$

7. $-90 \div 10$

8. $-64 \div (-16)$

9. $0 \div (-7)$

10. $-150 \div 30$

Solve

11. The temperature dropped 18°F in 6 hours at a steady rate. Change per hour?

12. A stock lost $48 over 4 days, the same each day. Daily change?

Exit Ticket

13. Why is $-5 \div 0$ undefined?

UNIT 5 · LESSON 5.4 · PRACTICE

Lesson 5.4: Negative Fractions and Signs of Quotients

Name _____ Date _____

I can explain why $-(\frac{p}{q}) = \frac{-p}{q} = \frac{p}{-q}$.

Warm-Up

1. $-18 \div 6$

2. $20 \div (-5)$

Simplify

3. $\frac{-24}{8}$

4. $\frac{35}{-7}$

5. $\frac{-9}{-3}$

6. $\frac{-5}{10}$

7. $\frac{6}{-8}$

8. $\frac{-15}{-20}$

Same or different?

9. $-\frac{2}{5}$ and $\frac{-2}{5}$

   A same

   B different

10. $\frac{-4}{-9}$ and $-\frac{4}{9}$

    A same

    B different

Exit Ticket

11. Write $-\frac{5}{6}$ two other ways.

WEEKLY CHECK · Fri Mar 26 · ABOUT 15 MINUTES

Weekly Check 13: Multiplying and Dividing Integers

Name _____ Date _____

Show what you know

1. $(-7) \times 8$

2. $(-6) \times (-9)$

3. $(-2)(-3)(-4)$

4. $-63 \div 7$

5. $-48 \div (-6)$

6. Simplify $\frac{14}{-21}$

UNIT 5 · LESSON 5.5 · PRACTICE

Lesson 5.5: Multiplying Signed Fractions

Name _____ Date _____

I can multiply positive and negative fractions and mixed numbers.

Warm-Up

1. $\frac{2}{3} \times \frac{3}{4}$

2. $(-2) \times 3$

Multiply

3. $(-\frac{1}{3}) \times \frac{6}{7}$

4. $\frac{4}{5} \times (-\frac{5}{8})$

5. $(-\frac{3}{4}) \times (-\frac{2}{9})$

6. $(-3\frac{1}{2}) \times \frac{4}{7}$

7. $(-2\frac{1}{2}) \times (-2\frac{1}{2})$

8. $3\frac{1}{3} \times (-\frac{9}{20})$

Solve

9. A diver descends at $2\frac{1}{4}$ ft per second for $\frac{4}{3}$ seconds. Change in depth?

Exit Ticket

10. $(-3\frac{3}{4}) \times (-1\frac{3}{5})$

UNIT 5 · LESSON 5.6 · PRACTICE

Lesson 5.6: Dividing Signed Fractions

Name _____ Date _____

I can divide positive and negative fractions.

Warm-Up

1. Reciprocal of $-\frac{3}{8}$

2. $\frac{3}{4} \times \frac{8}{3}$

Divide

3. $(-\frac{1}{2}) \div \frac{1}{4}$

4. $\frac{3}{5} \div (-\frac{9}{10})$

5. $(-\frac{4}{7}) \div (-\frac{2}{7})$

6. $(-6) \div \frac{3}{4}$

7. $1\frac{1}{4} \div (-2\frac{1}{2})$

8. $(-4\frac{1}{2}) \div (-\frac{3}{8})$

Solve

9. A hiker descends $4\frac{1}{2}$ miles in $\frac{3}{4}$ hour. Change in elevation per hour (in miles)?

Exit Ticket

10. $(-3\frac{1}{3}) \div \frac{5}{6}$

UNIT 5 · LESSON 5.7 · PRACTICE

Lesson 5.7: Multiplying and Dividing Signed Decimals

Name _____ Date _____

I can multiply and divide positive and negative decimals.

Warm-Up

1. 8×1.25

2. $10 \div 4$

Compute

3. $(-3.2) \times 5$

4. $(-0.25) \times (-12)$

5. $1.5 \times (-0.6)$

6. $(-8.4) \div 4$

7. $6.3 \div (-0.7)$

8. $(-0.48) \div (-0.06)$

Solve

9. A submarine changes depth by −12.5 m per minute. Change in 3.2 minutes?

10. A temperature fell 7.2°F in 1.5 hours. Change per hour?

Exit Ticket

11. $(-13.5) \div 0.45$

UNIT 5 · LESSON 5.8 · PRACTICE

Lesson 5.8: Rational Number Operations Practice

Name _____ Date _____

I can multiply and divide any rational numbers fluently.

Warm-Up

1. 0.75×0.8

2. $1.2 \div 1.5$

Compute

3. $(-0.25)(-\frac{4}{5})$

4. $-\frac{3}{8} \div 0.75$

5. $(-2.4)(\frac{5}{6})$

6. $-2\frac{1}{2} \div (-0.5)$

7. $(-1.5)(-1.5)$

8. $\frac{1}{3} \div (-\frac{1}{9})$

Exit Ticket

9. Evaluate $(-\frac{2}{5})(1.5) \div (-0.3)$.

WEEKLY CHECK · Fri Apr 2 · ABOUT 20 MINUTES

Weekly Check 14: Rational Products and Quotients

Name _____ Date _____

Show what you know

1. $(-\frac{3}{8}) \times \frac{4}{9}$

2. $(-2\frac{1}{2}) \times (-1\frac{1}{5})$

3. $(-\frac{4}{5}) \div \frac{2}{15}$

4. $2\frac{1}{3} \div (-1\frac{5}{9})$

5. $(-1.6) \times 0.5$

6. $(-2.7) \div (-0.09)$

UNIT 5 · LESSON 5.9 · PRACTICE

Lesson 5.9: Order of Operations with Rational Numbers

Name _____ Date _____

I can use the order of operations with positive and negative numbers.

Warm-Up

1. $(-2)^2$

2. $-3 + 2$

Evaluate

3. $5 - 2 \times (-4)$

4. $-18 \div 3 + (-2)$

5. $(-3)^2 - (-3)$

6. $-2(5 - 9)$

7. $(1 - 7)(-2) \div 4$

8. $-4^2 + (-4)^2$

Exit Ticket

9. Evaluate $-0.5(6 - 10) + 1.5^2$.

UNIT 5 · LESSON 5.10 · PRACTICE

Lesson 5.10: The Distributive Property with Rational Numbers

Name _____ Date _____

I can use the distributive property with negative numbers.

Warm-Up

1. $(-4) \times 3$

2. $(-4)(-0.25)$

Use the distributive property

3. -6×1.5 (think $6 \times (1 + 0.5)$)

4. $-8(10 - 0.5)$

5. $-3(4x + 2)$

6. $-(a - 9)$

7. $\frac{2}{3}(9 - 12)$

8. $-0.5(4n - 10)$

Exit Ticket

9. Show two ways to evaluate $-7(5 - 2)$.

UNIT 5 · LESSON 5.11 · PRACTICE

Lesson 5.11: Repeated Change: Rate × Time

Name _____ Date _____

I can model repeated change with multiplication of signed numbers.

Warm-Up

1. $(-4.5) \times 3$

2. $(-4.5) \times (-3)$

Solve

3. A tank drains 1.5 gallons per minute. Change after 8 minutes?

4. Same tank: change compared to 8 minutes ago?

5. Temperature rises 1.2° per hour. Compared to now, what was it 6 hours ago?

6. A diver ascends 0.75 m/s. Change in 20 s?

Exit Ticket

7. Explain $(-3)(-4) = 12$ using a temperature that drops 3° per hour.

UNIT 5 · LESSON 5.12 · PRACTICE

Lesson 5.12: Choosing the Operation

Name _____ Date _____

I can decide which operation fits a situation with rational numbers.

Warm-Up

1. $-1{,}240 \div 4$

2. $(-3.5) \times 6$

Which operation? Then solve.

3. A balance of −$45 gets a $60 deposit.

4. A plane descends 300 ft/min for 12 min.

5. Temperatures −6° and 9°: how far apart?

6. A stock drops $36.60 over 6 days. Average?

Exit Ticket

7. Write a situation for each operation using negative numbers.

WEEKLY CHECK · Fri Apr 9 · ABOUT 20 MINUTES

Weekly Check 15: Operations and Models

Name _____ Date _____

Show what you know

1. $-4 + 6 \times (-2)$

2. $(-5)^2 - 3(-4)$

3. $-3(2x - 7)$

4. Rate $-1.5°$/hr; 4 hours ago?

5. A $45 loss over 5 days: average daily change

6. From 30 to −10 in 8 hours: rate

UNIT 5 · LESSON 5.13 · PRACTICE

Lesson 5.13: Average Change

Name _____ Date _____

I can find the mean of positive and negative numbers.

Warm-Up

1. $-4 + 2 - 7 + 1 - 2$

2. $-10 \div 5$

Find the mean

3. $-6, 4, -2, 0$

4. $-10, -15, -5$

5. $3, -9, 6, -12, 2$

6. $-1, -1, -1, 7$

Solve

7. A stock changed −\$1.20, +\$0.60, −\$0.90, +\$0.30 over 4 days. Mean daily change?

Exit Ticket

8. Find 4 numbers with a mean of −3.

UNIT 5 · LESSON 5.14 · PRACTICE

Lesson 5.14: Rate Problems with Negatives

Name _____ Date _____

I can solve rate problems that involve decreases.

Warm-Up

1. $-6 - 18$

2. $-24 \div 8$

Solve

3. A submarine goes from −40 m to −220 m in 12 minutes. Rate?

4. At that rate, how long from −220 m to −400 m?

5. A plant's height changes from 14.5 cm to 11.3 cm in 4 days (drought). Rate?

6. −$2.40 per day. After how many days is the change −$16.80?

Exit Ticket

7. Temperature drops 2.5° per hour from 10°. What is it after 6 hours?

UNIT 5 · LESSON 5.15 · PRACTICE

Lesson 5.15: Multi-Step Rational Number Problems

Name _____ Date _____

I can solve multi-step problems using all four operations with rational numbers.

Warm-Up

1. $3 \times (-1{,}250)$

2. $2 \times 2{,}400$

Solve

3. A diver descends 2.4 m every 10 seconds for 50 seconds, then rises 3 m. Final change?

4. Ellie's savings change by −$12.50 a week for 4 weeks, then +$20 a week for 3 weeks. Net change?

5. The temperature was 4°C. It fell 1.5°C per hour for 6 hours, then rose 2°C. Final temperature?

Exit Ticket

6. Write a problem whose answer is −8 using all four operations.

UNIT 5 · LESSON 5.16 · PRACTICE

Lesson 5.16: Rational Number Operations Problem Solving

Name _____ Date _____

I can use all four operations with rational numbers to solve problems.

Warm-Up

1 $4 \times (-3)$

2 $5 + (-12)$

Mixed practice

3 $(-8) \times 7$

4 $-96 \div (-12)$

5 $(-\frac{5}{6}) \times \frac{9}{10}$

6 $(-1\frac{3}{4}) \div 3\frac{1}{2}$

7 $(-0.4) \times (-2.75)$

8 $-3 + (-2)^3$

Solve

9 A stock's price changes −$0.85 per day for 6 days, then +$1.20 per day for 4 days. Net change? Average per day?

Exit Ticket

10 Explain how you decide the sign of a product of five numbers.

UNIT REVIEW · Thu Apr 15 (homework) or as needed · ABOUT 35 MINUTES

Unit 5 Review

Name _____ Date _____

Compute

1. $(-12) \times (-5)$

2. $84 \div (-7)$

3. $(-\frac{2}{3}) \times (-\frac{9}{14})$

4. $(-\frac{5}{8}) \div \frac{15}{16}$

5. $3.5 \times (-0.6)$

6. $(-4.5) \div 1.5$

7. $(-1)^6$

8. $10 - 3(-2)^2$

Apply

9. Mean of −8, 3, −5, −2

10. Temp falls from 12° to −9° in 7 hours. Rate?

11. $-0.5(8x - 6)$

12. A pool drains 2.5 in./hour. Change after 3.5 hours?

UNIT TEST · Fri Apr 16 · ABOUT 45 MINUTES

Unit 5 Test

Name _____ Date _____

Show your work. This test has 21 points.

Part A · Integers (1 point each)

1. $(-9) \times 6$

2. $(-11) \times (-4)$

3. $(-1)(-2)(-3)(-4)$

4. $-108 \div 9$

5. $-75 \div (-15)$

6. Simplify $^{-16}/_{24}$

Part B · Rational numbers (1 point each)

7. $(-^{4}/_{7}) \times 2\,^{5}/_{8}$

8. $(-^{3}/_{5}) \div (-^{9}/_{20})$

9. $(-2.4) \times 1.25$

10. $6.4 \div (-0.8)$

11. $-8 + 12 \div (-4)$

12. $-4(3 - 2.5)$

Part C · Applications (2 points each)

13. The temperature fell from 15°F to −9°F in 6 hours. Find the rate of change.

14. Mean of −3.5, 2, −4.5, 1, −5

15. A submarine changes depth by −8.5 m per minute. Where was it, relative to now, 4 minutes ago?

Part D · Explain (3 points)

16 Use a pattern or the distributive property to explain why (−2)(−5) = 10.

UNIT 6

Expressions, Equations, and Inequalities

Writing and simplifying expressions, solving two-step equations, and solving and graphing inequalities.

Your practice pages for Unit 6. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

variable	A letter for a number.
expression	No equals sign.
equation	Two expressions are equal.
inequality	Comparison with $<$, $>$, $\leq$, $\geq$.
like terms	Same variable part.

UNIT 6 · LESSON 6.1 · PRACTICE

Lesson 6.1: Writing Expressions

Name _____ Date _____

I can write algebraic expressions from words and real situations.

Warm-Up

1. $9 \times 4 + 6$

2. Name the coefficient in $-5y$

Write an expression

3. a number increased by 12

4. 8 times a number, minus 4

5. the difference of a number and −2

6. one-third of a number

7. 6 less than 5 times a number

8. twice the sum of a number and 3

Write an expression for the situation

9. A phone plan costs $20 plus $0.10 per text t.

10. Lillie has d dollars and spends $7.50.

Exit Ticket

11. Write a real situation for the expression $12h + 15$.

UNIT 6 · LESSON 6.2 · PRACTICE

Lesson 6.2: Evaluating Expressions

Name _____ Date _____

I can evaluate expressions for given values, including negative and fractional values.

Warm-Up

1. $1.8 \times (-10)$

2. $-18 + 32$

Evaluate when $x = -3$, $y = 4$, $z = 0.5$

3. $2x + y$

4. $y - 5x$

5. $10z - x$

6. xy

7. $x^2 + y^2$

8. $(y - x) \div z$

Solve

9. A taxi costs $2.5m + 3$ dollars for m miles. Cost for 12 miles?

10. Convert 25°C to Fahrenheit with $1.8C + 32$.

Exit Ticket

11. Evaluate $-2(x - 6)$ when $x = -1$.

UNIT 6 · LESSON 6.3 · PRACTICE

Lesson 6.3: Combining Like Terms

Name _____ Date _____

I can simplify expressions by combining like terms, including rational coefficients.

Warm-Up

1. $4 - 6$

2. $0.5 - 0.75$

Simplify

3. $9x + 4x$

4. $3a - 8 + 5a$

5. $-6n + 2 - n + 9$

6. $2.4p - 1.4p + 6$

7. $\frac{2}{3}k + \frac{1}{3}k - 5$

8. $4x + 3y - x - 5y$

Exit Ticket

9. Is $5x + 3$ equal to $8x$? Explain.

UNIT 6 · LESSON 6.4 · PRACTICE

Lesson 6.4: Expressions in Context

Name _____ Date _____

I can rewrite an expression to show what it means in a situation.

Warm-Up

1 $1 + 0.05$

2 $1 - 0.2$

Rewrite and explain

3 $p - 0.35p$

4 $c + 0.07c$

5 $2w + 2(w + 4)$

6 $x + x + 0.5x$

Exit Ticket

7 Write two expressions for the price p after a 10% increase. Explain each.

WEEKLY CHECK · Fri Apr 23 · ABOUT 15 MINUTES

Weekly Check 16: Expressions

Name _____ Date _____

Show what you know

1. Write: 4 less than 3 times a number

2. Evaluate $5x - 2y$ when $x = -2$, $y = 3$

3. Simplify $7a - 3 - 9a + 8$

4. Simplify $0.6k + 1.4k - 2$

5. Rewrite $p - 0.25p$

6. Evaluate $x^2 - x$ when $x = -4$

UNIT 6 · LESSON 6.5 · PRACTICE

Lesson 6.5: The Distributive Property: Expanding

Name _____ Date _____

I can expand expressions using the distributive property.

Warm-Up

1. $-2 \cdot (-5)$

2. $\frac{3}{4} \cdot 12$

Expand

3. $3(x + 7)$

4. $-4(2y - 1)$

5. $0.5(6k + 8)$

6. $-(-3 + p)$

7. $\frac{2}{3}(9x - 3)$

8. $-1.5(4 - 2w)$

Expand and simplify

9. $2(x - 4) + 3x$

10. $6 - 2(n + 5)$

11. $3(a + 1) - (a - 4)$

12. $-5(2 - y) + 4y$

Exit Ticket

13. Expand and simplify: $-3(2x + 4) + 7x$

UNIT 6 · LESSON 6.6 · PRACTICE

Lesson 6.6: Factoring Expressions

Name _____ Date _____

I can factor an expression by finding a common factor.

Warm-Up

1. GCF of 6 and 9

2. Expand 3(2x + 3)

Factor

3. 4x + 20

4. 18y − 24

5. 9k + 9

6. −5p + 30

7. 12 − 16w

8. 2.5x + 5

Exit Ticket

9. A rectangle's area is 8x + 32 and one side is 8. Find the other side.

UNIT 6 · LESSON 6.7 · PRACTICE

Lesson 6.7: Simplifying with Rational Coefficients

Name _____ Date _____

I can simplify expressions with fraction and decimal coefficients.

Warm-Up

1. $\frac{1}{2} \cdot 4x$

2. $0.4 \cdot 5n$

Simplify

3. $\frac{1}{3}(6x + 3) - x$

4. $0.5(4k - 8) + 3k$

5. $-\frac{3}{4}(4p - 8) + p$

6. $1.2(5 + 10w) - 2w$

7. $\frac{1}{2}x + \frac{1}{4}x - \frac{3}{4}x$

8. $\frac{2}{3}(3n - 6) - \frac{1}{2}(2n + 4)$

Exit Ticket

9. Simplify $\frac{3}{5}(10x - 5) - 0.5(2x + 4)$

UNIT 6 · LESSON 6.8 · PRACTICE

Lesson 6.8: Equivalent Expressions

Name _____ Date _____

I can tell whether two expressions are equivalent and justify my answer.

Warm-Up

1 Expand 2(x + 3)

2 3(1) + 2

Equivalent? Justify.

3 6x − 9 and 3(2x − 3)

4 −(x − 4) and −x − 4

5 0.5(4n + 2) and 2n + 1

6 3(a + 2) − a and 2a + 2

Exit Ticket

7 Write three expressions equivalent to 4x + 8.

WEEKLY CHECK · Fri Apr 30 · ABOUT 15 MINUTES

Weekly Check 17: Equivalent Expressions

Name _____ Date _____

Show what you know

1. Expand $-3(4x - 2)$

2. Factor $12y + 18$

3. Simplify $\frac{1}{2}(6n + 4) - n$

4. Simplify $5(x - 2) - 2(x + 1)$

5. Equivalent? $2(x + 5)$ and $2x + 5$

6. Factor $-8m + 24$

UNIT 6 · LESSON 6.9 · PRACTICE

Lesson 6.9: One-Step Equations

Name _____ Date _____

I can solve one-step equations with rational numbers.

Warm-Up

1. $-2 + 4.5$

2. $18 \div (-3)$

Solve

3. $a + 12 = 5$

4. $b - 3.2 = -1.8$

5. $6c = -45$

6. $-d = 9$

7. $p \div (-2) = 11$

8. $\frac{3}{4}w = -9$

Exit Ticket

9. Solve and check: $-\frac{1}{2}x = 7$.

UNIT 6 · LESSON 6.10 · PRACTICE

Lesson 6.10: Two-Step Equations

Name _____ Date _____

I can solve equations of the form $px + q = r$.

Warm-Up

1. 115 − 25

2. 90 ÷ 15

Solve

3. $2x + 7 = 19$

4. $6x - 4 = 32$

5. $-3x + 10 = 1$

6. $4x + 15 = 3$

7. $-7x - 2 = 26$

8. $8x + 5 = 9$

Fraction and decimal coefficients

9. $\frac{1}{2}x + 4 = 10$

10. $0.25x - 3 = 2$

11. $\frac{2}{3}x - 5 = -1$

12. $1.5x + 2.5 = -5.5$

Exit Ticket

13. Write and solve: a taxi charges $3 plus $2.50 per mile. The fare was $20.50. How many miles?

UNIT 6 · LESSON 6.11 · PRACTICE

Lesson 6.11: Equations with Parentheses

Name _____ Date _____

I can solve equations of the form $p(x + q) = r$.

Warm-Up

1. $33 \div 3$

2. $11 - 6$

Solve

3. $4(x + 3) = 28$

4. $-2(y - 7) = 6$

5. $6(k - 1.5) = 21$

6. $-3(m + 4) = -3$

7. $\frac{2}{3}(p - 9) = 4$

8. $0.5(w + 8) = -1$

Exit Ticket

9. The perimeter of a rectangle is $2(L + 7) = 46$. Find L.

UNIT 6 · LESSON 6.12 · PRACTICE

Lesson 6.12: Equation Word Problems

Name _____ Date _____

I can write and solve equations for word problems and compare to an arithmetic solution.

Warm-Up

1 $2x + 12 = 54$

2 $54 - 12$

Write an equation and solve

3 A family pays $28 for 4 movie tickets and a $4 snack. Price per ticket?

4 A water tank has 120 gallons and drains 7.5 gallons per minute. When will it have 30 gallons left?

5 Ellie's age times 3, minus 4, is 32. How old is she?

6 A temperature of −8°F rises 2.5° per hour. When will it reach 7°F?

Exit Ticket

7 Write your own equation word problem with answer $x = -3$.

WEEKLY CHECK · Fri May 7 · ABOUT 20 MINUTES

Weekly Check 18: Equations

Name _____ Date _____

Show what you know

1. $x - 6.5 = -2$

2. $-7y = 56$

3. $3x - 8 = 13$

4. $-5x + 4 = 29$

5. $4(n - 2) = -20$

6. A $15 fee plus $9 per hour; total $78. Hours?

UNIT 6 · LESSON 6.13 · PRACTICE

Lesson 6.13: Writing and Graphing Inequalities

Name _____ Date _____

I can write inequalities for situations and graph them on a number line.

Warm-Up

1. Is $48 \geq 48$ true?

2. Is $48 > 48$ true?

Write an inequality

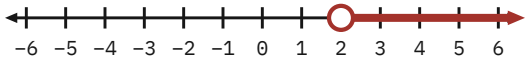
3. a speed limit of 65 mph

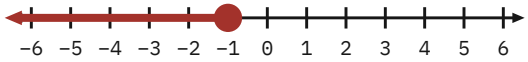
4. more than 12 players

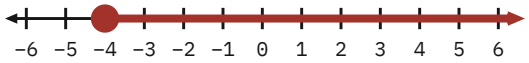
5. at least $20

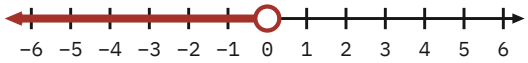
6. colder than −10°

Write the inequality shown

7. 

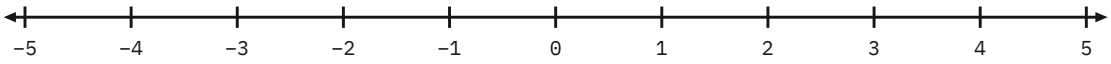
8. 

9. 

10. 

Exit Ticket

11. Graph $t \leq 3$ on a number line.



UNIT 6 · LESSON 6.14 · PRACTICE

Lesson 6.14: Solving One-Step Inequalities

Name _____ Date _____

I can solve one-step inequalities, including multiplying or dividing by a negative.

Warm-Up

1. Is $-2 > -5$?

2. $12 \div (-3)$

Solve

3. $x - 4 < 7$

4. $4k \geq -20$

5. $-6m > 18$

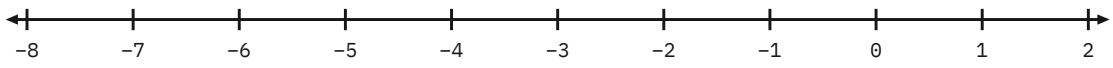
6. $-p \leq 2.5$

7. $\frac{1}{3}w < -2$

8. $-0.5z \geq -4$

Exit Ticket

9. Solve $-2x \geq 10$ and graph it.



UNIT 6 · LESSON 6.15 · PRACTICE

Lesson 6.15: Two-Step Inequalities

Name _____ Date _____

I can solve inequalities of the form $px + q > r$ and graph the solutions.

Warm-Up

1. 100 − 20

2. 80 ÷ 8

Solve

3. $2x + 7 > 15$

4. $5y - 3 \leq -18$

5. $-3k + 1 \geq 10$

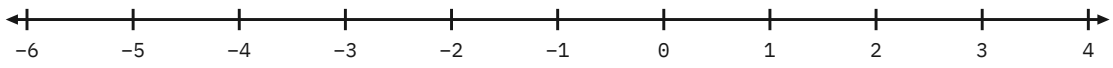
6. $9 - 2m < 1$

7. $0.5p + 4 > 1$

8. $-x - 6 \leq -2$

Exit Ticket

9. Solve $6 - 3x \leq 12$ and graph it.



UNIT 6 · LESSON 6.16 · PRACTICE

Lesson 6.16: Inequality Word Problems

Name _____ Date _____

I can write, solve, and interpret inequalities for real situations.

Warm-Up

1 150 − 10

2 140 ÷ 12.5

Write, solve, and interpret

3 A taxi charges $3 plus $2 per mile. You have $25. How far can you ride?

4 To keep a B, Lillie needs an average of at least 80 on 4 tests. Her first three scores are 76, 84, 78. What score does she need on test 4?

5 A freezer is at 5°F and cools 3° per hour. When will it be below −10°F?

Exit Ticket

6 Write an inequality problem where the answer must be a whole number. Solve it.

UNIT REVIEW · Thu May 13 (homework) or as needed · ABOUT 40 MINUTES

Unit 6 Review

Name _____ Date _____

Expressions

1. Evaluate $-2a + b^2$ when $a = 3$, $b = -4$

2. Simplify $3(2x - 1) - (x + 5)$

3. Factor $15k - 25$

4. Rewrite $1.08c$ in words

Equations and inequalities

5. $4x + 9 = -11$

6. $\frac{2}{5}(x + 10) = 6$

7. $-3x + 4 > 19$

8. $7 + 2y \leq 1$

9. A pool has 900 gal and drains 25 gal/min. When is it below 400 gal?

UNIT TEST · Fri May 14 · ABOUT 45 MINUTES

Unit 6 Test

Name _____ Date _____

Show your work. This test has 21 points.

Part A · Expressions (1 point each)

1. Write: twice the difference of a number and 5

2. Evaluate $3x - y$ when $x = -2$, $y = -7$

3. Simplify $-4m + 9 + 6m - 12$

4. Expand $-5(2 - 3k)$

5. Factor $21a + 14$

6. Simplify $\frac{3}{4}(8x - 4) - 2x$

Part B · Equations (1 point each)

7. $y + 8.25 = 3$

8. $-6x + 0 = 39$

9. $2x + 11 = -5$

10. $-3(p + 4) = 9$

11. $\frac{1}{3}n - 2 = 5$

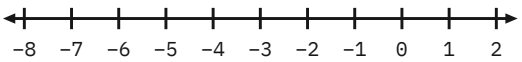
Part C · Inequalities (1 point each)

12. $x - 3 \geq -8$

13. $-4y < 28$

14. $5 - 2k \leq 17$

15. Graph the answer to the previous problem.



Part D · Problem solving (3 points each)

16 A plumber charges $55 plus $40 per hour. The bill was $235. Write and solve an equation to find the hours.

17 Ellie has $35 and wants to buy shirts at $8.50 each with a $3 shipping fee. Write and solve an inequality for the number of shirts. Interpret.

UNIT 7

Statistics and Probability

Sampling, comparing populations, probability models, and compound events.

Your practice pages for Unit 7. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

population	The whole group.
sample	Part of the population.
MAD	Average distance from the mean.
probability	A number 0–1 for likelihood.
sample space	All possible outcomes.

UNIT 7 · LESSON 7.1 · PRACTICE

Lesson 7.1: Populations and Samples

Name _____ Date _____

I can tell the difference between a population and a sample.

Warm-Up

1 What fraction of 2,000 is 80?

2 What percent of 2,000 is 80?

Name the population and the sample

3 A city surveys 400 residents about a new park.

4 A coach times 6 of her 30 swimmers.

5 A scientist tags 25 frogs in a pond.

6 A teacher checks every student's homework.

Exit Ticket

7 Describe a question you could answer with a sample. Name the population and the sample.

UNIT 7 · LESSON 7.2 · PRACTICE

Lesson 7.2: Random Samples and Bias

Name _____ Date _____

I can tell whether a sample is random and representative, or biased.

Warm-Up

1 Name a population for "favorite sport in town."

2 Name a sample of that population.

Random or biased? Explain.

3 A school picks 40 student ID numbers by computer to survey about lunch.

4 A pet store asks its customers whether people in the city prefer cats or dogs.

5 A magazine asks readers to mail in opinions about the magazine.

6 A farmer tests corn from 10 spots chosen at random across the field.

Exit Ticket

7 Describe a random way to choose 10 students from a school of 300.

UNIT 7 · LESSON 7.3 · PRACTICE

Lesson 7.3: Making Inferences from Samples

Name _____ Date _____

I can use data from a random sample to estimate something about a population.

Warm-Up

1. 18 ÷ 50 as a percent

2. 0.36 × 600

Estimate for the population

3. 9 of 30 in the sample; population 450

4. 14 of 70 in the sample; population 1,500

5. 6 of 25 in the sample; population 800

6. 33 of 120 in the sample; population 2,400

Solve

7. A random sample of 80 households found 52 own a pet. The town has 6,000 households. Estimate the number with pets.

8. A survey of 25 random students found 7 play an instrument. Estimate for 350 students.

Exit Ticket

9. Why should the sample be random for this estimate to be trusted?

UNIT 7 · LESSON 7.4 · PRACTICE

Lesson 7.4: Variability in Samples

Name _____ Date _____

I can use several samples to judge how much an estimate might vary.

Warm-Up

1 Mean of 6, 8, 5, 7, 9

2 7 out of 20 as a percent

Use the samples

3 Four random samples of 50 voters show 28, 31, 25, and 30 supporting a plan. Find the mean percent.

4 Use your answer to estimate supporters among 8,000 voters.

5 Lillie's sample of 10 says 70% like a book. Ellie's sample of 100 says 52%. Whose estimate is more reliable? Why?

Exit Ticket

6 Why don't all random samples give the same result?

WEEKLY CHECK · Fri May 21 · ABOUT 15 MINUTES

Weekly Check 19: Sampling

Name _____ Date _____

Show what you know

1. A school surveys 60 of its 900 students. Name the population.

2. Surveying only the chess club about favorite games: random or biased?

3. 15 of 60 sampled students bike. Estimate bikers in 900.

4. Samples show 40%, 46%, 43%, 39%. Mean percent?

5. Use that mean to estimate for 2,000 people.

6. Which is more reliable: a sample of 30 or of 300?

UNIT 7 · LESSON 7.5 · PRACTICE

Lesson 7.5: Measures of Center and Variability

Name _____ Date _____

I can find the mean, median, mean absolute deviation, range, and interquartile range of a data set.

Warm-Up

1. Order: 9, 2, 7, 4, 5

2. Median of 2, 4, 5, 7, 9

Find the mean and MAD

3. 4, 6, 8, 10, 12

4. 5, 5, 7, 9, 9

5. 10, 12, 14, 16, 18, 20

6. 1, 4, 4, 5, 6

Find the median and IQR

7. 2, 3, 5, 6, 8, 9, 11

8. 10, 12, 13, 15, 18, 20, 21, 25

Exit Ticket

9. Which set has more variability: 2, 4, 6, 8, 10 or 12, 15, 15, 18, 20? Use MAD.

UNIT 7 · LESSON 7.6 · PRACTICE

Lesson 7.6: Comparing Dot Plots

Name _____ Date _____

I can compare two data sets using their centers and variability.

Warm-Up

1 Mean of 3, 4, 4, 5, 5, 5, 6, 6, 7

2 Mean of 6, 7, 7, 8, 8, 8, 9, 9, 10

Compare the data sets

3 Team X jumps (ft): 10, 11, 12, 12, 13, 14. Team Y: 12, 13, 14, 14, 15, 16. Difference of means?

4 Find the MAD of either team.

5 How many MADs apart are the means?

6 Is the difference large or small? Explain.

Exit Ticket

7 Two data sets have means 20 and 22 and MADs of 5. Are they very different? Why?

UNIT 7 · LESSON 7.7 · PRACTICE

Lesson 7.7: Comparing Box Plots

Name _____ Date _____

I can compare two data sets displayed in box plots.

Warm-Up

1 What does the line inside a box show?

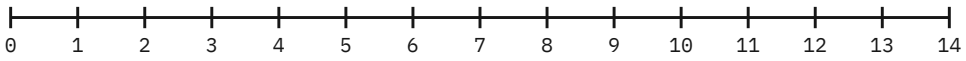
2 What does the box's length show?

Compare

3 Data: 3, 5, 6, 8, 9, 11, 14. Five-number summary?

4 Data: 20, 22, 25, 26, 28, 30, 31, 35. Five-number summary?

5 Plot Set A {2, 4, 5, 7, 9} and Set B {5, 7, 8, 10, 12}. Which has the greater median?



Exit Ticket

6 When might you compare medians instead of means?

UNIT 7 · LESSON 7.8 · PRACTICE

Lesson 7.8: Comparing Populations

Name _____ Date _____

I can use random samples to compare two populations.

Warm-Up

1 Mean of 4, 5, 6, 6, 7, 8

2 Mean of 2, 3, 3, 4, 4, 5

Compare the populations

3 Random samples of tomato plants: Fertilizer A heights (cm) have mean 42, MAD 3. Fertilizer B: mean 36, MAD 3. How many MADs apart?

4 Is there a meaningful difference? Explain.

5 Samples of two dog breeds: mean weights 61 lb and 58 lb, each MAD 6 lb. Meaningful difference?

Exit Ticket

6 Write a question comparing two populations you could answer with random samples.

WEEKLY CHECK · Fri May 28 · ABOUT 15 MINUTES

Weekly Check 20: Comparing Data

Name _____ Date _____

Show what you know

1. Mean of 6, 8, 10, 12, 14

2. MAD of 6, 8, 10, 12, 14

3. Median of 3, 7, 8, 12, 15, 20

4. IQR of 2, 4, 5, 7, 8, 10, 13, 14

5. Means 30 and 36, MAD 3. How many MADs apart?

6. Is that a meaningful difference?

UNIT 7 · LESSON 7.9 · PRACTICE

Lesson 7.9: Probability and Likelihood

Name _____ Date _____

I can describe the likelihood of an event with a number from 0 to 1.

Warm-Up

1. Is $\frac{3}{4}$ closer to 0 or 1?

2. Write 0.2 as a fraction

Describe the likelihood

3. Probability 0.95

4. Probability $\frac{1}{8}$

5. Probability 0

6. Probability 0.5

7. Snow in July in Texas

8. Choosing a vowel from A, E, I

Exit Ticket

9. Describe an event that is likely but not certain. Estimate its probability.

UNIT 7 · LESSON 7.10 · PRACTICE

Lesson 7.10: Theoretical Probability

Name _____ Date _____

I can find the theoretical probability of an event.

Warm-Up

1. Simplify $^{5}/_{10}$

2. $1 - 0.3$

A spinner has 8 equal parts numbered 1–8. Find each probability.

3. P(3)

4. P(even)

5. P(less than 4)

6. P(9)

7. P(not 8)

8. P(multiple of 3)

Solve

9. A class has 12 boys and 16 girls. A name is drawn at random. P(girl)?

10. The probability of rain is 0.35. What is the probability of no rain?

Exit Ticket

11. Letters of MISSISSIPPI are on cards. P(S)?

UNIT 7 · LESSON 7.11 · PRACTICE

Lesson 7.11: Experimental Probability

Name _____ Date _____

I can find experimental probability and compare it to theoretical probability.

Warm-Up

1. 15 ÷ 60 as a percent

2. $\frac{1}{6}$ of 600

A coin is flipped 50 times: 29 heads.

3. Experimental P(heads)

4. Theoretical P(heads)

5. Experimental P(tails)

6. Predict heads in 1,000 flips of a fair coin

Predict

7. A player made 18 of 24 free throws. Predict makes in 40 attempts.

8. A spinner landed on blue 12 times in 80 spins. Predict blues in 200 spins.

Exit Ticket

9. Why might experimental and theoretical probability differ?

UNIT 7 · LESSON 7.12 · PRACTICE

Lesson 7.12: Probability Models

Name _____ Date _____

I can build uniform and non-uniform probability models.

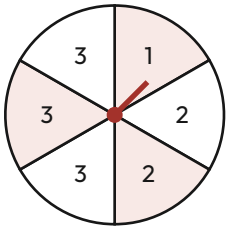
Warm-Up

1 $\frac{1}{2} + \frac{1}{4} + \frac{1}{4}$

2 $1 - 0.3 - 0.5$

Build or check the model

3 Spinner with sections 1, 2, 2, 3, 3, 3. Give P(1), P(2), P(3).



4 Is the model P(A) = 0.4, P(B) = 0.4, P(C) = 0.3 valid? Why?

5 Of 50 cars, 20 were white, 15 black, 10 silver, and 5 red. Write a model.

Exit Ticket

6 Design a spinner where P(green) = 0.5 and P(red) = 0.25.

WEEKLY CHECK · Fri Jun 4 · ABOUT 15 MINUTES

Weekly Check 21: Probability

Name _____ Date _____

Show what you know

1. Bag: 4 red, 6 blue. P(red)

2. P(not 2) on a number cube

3. Probability 0.9 is...

4. 32 heads in 50 flips: experimental P(heads)

5. Model: 0.25, 0.45, ? Find ?

6. Coin landed heads 12 of 40 times. Predict heads in 200 flips with that rate.

UNIT 7 · LESSON 7.13 · PRACTICE

Lesson 7.13: Sample Spaces for Compound Events

Name _____ Date _____

I can list the sample space of a compound event using lists and tables.

Warm-Up

1. 6×6

2. Simplify $\frac{6}{36}$

Use the sum table

3. P(sum is 8)

4. P(sum greater than 9)

5. P(sum is even)

6. P(sum less than 5)

7. P(sum is 12)

8. P(sum is 1)

List the sample space

9. Flip a coin and spin a 3-color spinner (R, B, Y). List all outcomes.

Exit Ticket

10. Which sum of two number cubes is most likely? Why?

UNIT 7 · LESSON 7.14 · PRACTICE

Lesson 7.14: Tree Diagrams

Name _____ Date _____

I can use a tree diagram to find the sample space of a compound event.

Warm-Up

1 3×2

2 $2 \times 2 \times 2$

Tree diagrams

3 Draw a tree: flip a coin twice. List outcomes.

4 P(exactly one head)

5 P(at least one tail)

Count

6 4 sandwiches, 3 drinks: combos

7 2 sizes, 5 flavors: cones

8 Coin and number cube: outcomes

9 3 coins: P(all heads)

Exit Ticket

10 A café offers 3 breads, 4 fillings, and 2 sauces. How many sandwiches?

UNIT 7 · LESSON 7.15 · PRACTICE

Lesson 7.15: Probability of Compound Events

Name _____ Date _____

I can find the probability of a compound event using the sample space.

Warm-Up

1. $\frac{1}{2} \times \frac{1}{3}$

2. 2×6

Find each probability

3. Two coins: P(HH)

4. Two number cubes: P(both 6)

5. Two cubes: P(both even)

6. Coin + cube: P(T and 1 or 2)

7. Spinner 1–4 twice: P(sum 5)

8. Three coins: P(exactly 2 heads)

Solve

9. Ellie guesses on 2 true/false questions. P(both correct)?

10. A family has 3 children. Assuming boy and girl are equally likely, P(at least one girl)?

Exit Ticket

11. Roll two number cubes. P(sum is 4)?

UNIT 7 · LESSON 7.16 · PRACTICE

Lesson 7.16: Simulations

Name _____ Date _____

I can design and use a simulation to estimate the probability of a compound event.

Warm-Up

1 Mean of 6, 5, 7, 5, 7

2 What tool has 4 equal outcomes?

Design a simulation

3 A player makes 50% of shots. Design a simulation for P(making 3 in a row).

4 Rain is forecast at 30% each day. Design a simulation for P(rain on at least 1 of 2 days).

5 In 20 trials of the shot simulation, 3 had three heads. Experimental probability?

6 What is the theoretical probability of three heads?

Exit Ticket

7 Why do more trials give a better estimate?

UNIT REVIEW · Thu Jun 10 (homework) or as needed · ABOUT 40 MINUTES

Unit 7 Review

Name _____ Date _____

Statistics

1. Random or biased: surveying shoppers at a bookstore about reading habits

2. 24 of 80 sampled like a show. Estimate for 5,000.

3. Mean and MAD of 5, 7, 9, 11, 13

4. Five-number summary of 1, 4, 6, 7, 9, 12, 14

Probability

5. Spinner 1–10: P(prime)

6. Two coins: P(no heads)

7. Two cubes: P(sum is 6)

8. 3 shirts, 4 pants, 2 shoes: outfits

9. Design a simulation for a 25% event.

UNIT TEST · Fri Jun 11 · ABOUT 45 MINUTES

Unit 7 Test

Name _____ Date _____

Show your work. This test has 20 points.

Part A · Sampling (1 point each)

1. A factory tests 100 of 20,000 parts. Name the sample.

2. Asking the first 20 people in line for a roller coaster about favorite rides: random or biased?

3. 9 of 45 sampled students have a summer job. Estimate for 1,200 students.

4. Samples: 12, 15, 9, 14 successes out of 30 each. Mean count?

Part B · Comparing data (1 point each)

5. Mean of 8, 10, 11, 13, 18

6. MAD of 8, 10, 11, 13, 18

7. IQR of 5, 6, 9, 10, 12, 15, 16, 20

8. Means 52 and 60, MADs 4. MADs apart?

Part C · Probability (1 point each)

9. Bag: 2 red, 3 green, 5 yellow. P(green)

10. P(not yellow)

11. Coin + cube: P(H and 3)

12. Two cubes: P(sum is 9)

13. Three coins: P(at least one head)

14. 18 of 30 spins landed on red. Experimental P(red)?

Part D · Problem solving (3 points each)

15 Random samples of two orchards: Orchard A apples mean 150 g, MAD 10 g; Orchard B mean 175 g, MAD 10 g. Compare the orchards. Is the difference meaningful? Explain.

16 Design a simulation to estimate the probability that a family with 4 children has at least 3 girls. Describe the tool, one trial, and what you record.

UNIT 8

Scale Drawings and Geometric Figures

Scaled copies, scale drawings and maps, drawing triangles from conditions, and cross sections.

Your practice pages for Unit 8. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

scale factor	Multiplier for every length.
scale	Drawing length : actual length.
triangle inequality	Two sides sum to more than the third.
cross section	The shape of a slice.

UNIT 8 · LESSON 8.1 · PRACTICE

Lesson 8.1: Scaled Copies

Name _____ Date _____

I can tell whether one figure is a scaled copy of another.

Warm-Up

1. $6 \div 3$

2. $8 \div 4$

Scaled copy? Write yes or no.

3. 2×3 and 6×9

4. 4×6 and 6×8

5. 5×10 and 1×2

6. 3×7 and 9×14

7. 8×12 and 2×3

8. 1×4 and 3×7

Draw a scaled copy of the L-shape with every length × 3.

9.

Exit Ticket

10 Lillie adds 2 to every side of a triangle. Is the new triangle a scaled copy? Explain.

UNIT 8 · LESSON 8.2 · PRACTICE

Lesson 8.2: Scale Factors

Name _____ Date _____

I can find and use a scale factor.

Warm-Up

1. $10 \div 4$

2. 6×2.5

Find the scale factor

3. $3 \to 12$

4. $10 \to 4$

5. $6 \to 9$

6. $20 \to 5$

7. $8 \to 8$

8. $4 \to 14$

Find the missing length

9. Original 6 cm, scale factor 3

10. Original 12 cm, scale factor $\frac{1}{3}$

11. Original 9 cm, scale factor $1\frac{2}{3}$

12. Original 14 cm, scale factor 0.5

13. Original 10 cm, scale factor 1.5

14. Original 16 cm, scale factor 0.75

Exit Ticket

15. A triangle has sides 6, 8, 10. A copy's shortest side is 9. Find the other two sides.

UNIT 8 · LESSON 8.3 · PRACTICE

Lesson 8.3: Scale Factors and Area

Name _____ Date _____

I can find how area changes when a figure is scaled.

Warm-Up

1 2^2

2 $(\frac{1}{2})^2$

Find the new area

3 Area 8, scale factor 2

4 Area 6, scale factor 5

5 Area 50, scale factor 0.2

6 Area 12, scale factor 1.5

7 Area 20, scale factor 0.5

8 Area 3, scale factor 10

Solve

9 A garden 4 m by 5 m is redesigned with every length × 1.5. Find the new area.

10 A poster's area is 9 times a flyer's area. What is the scale factor?

Exit Ticket

11 Lillie says scaling by 3 triples the area. Explain her mistake.

UNIT 8 · LESSON 8.4 · PRACTICE

Lesson 8.4: Scale Drawings

Name _____ Date _____

I can use a scale to find actual lengths from a scale drawing.

Warm-Up

1. 1.5 × 8

2. 1.75 × 8

Find the actual length

3. Scale 1 in. : 6 ft; drawing 4 in.

4. Scale 1 cm : 3 m; drawing 8.5 cm

5. Scale 1 in. : 50 mi; drawing 2.25 in.

6. Scale 1 cm : 10 km; drawing 6.3 cm

7. Scale 2 in. : 5 ft; drawing 6 in.

8. Scale $1/4$ in. : 1 ft; drawing 3 in.

Exit Ticket

9. A model car uses 1 in. : 2 ft. The model is 7 in. long. How long is the real car?

WEEKLY CHECK · Fri Jun 18 · ABOUT 15 MINUTES

Weekly Check 22: Scale Factors

Name _____ Date _____

Show what you know

1. Is 6 × 15 a scaled copy of 2 × 5?

2. Original 8 cm, copy 20 cm. Scale factor?

3. Side 12 in., scale factor $2/3$

4. Area 7, scale factor 3. New area?

5. Scale 1 in. : 5 ft; drawing 6.4 in. Actual?

6. Scale 1 cm : 4 m; drawing 3.5 cm. Actual?

UNIT 8 · LESSON 8.5 · PRACTICE

Lesson 8.5: Finding Drawing Lengths

Name _____ Date _____

I can find the length on a scale drawing for a given actual length.

Warm-Up

1. $100 \div 5$

2. $10 \times \frac{1}{2}$

Find the drawing length

3. Scale 1 in. : 10 ft; actual 45 ft

4. Scale 1 cm : 2 m; actual 15 m

5. Scale 1 in. : 25 mi; actual 175 mi

6. Scale 1 cm : 50 km; actual 325 km

7. Scale 3 cm : 1 m; actual 4 m

8. Scale $\frac{1}{4}$ in. : 1 ft; actual 18 ft

Plan a room

9. Lillie's room is 12 ft by 10 ft. Draw it using the scale 1 square : 2 ft. What are the drawing dimensions?

Exit Ticket

10. A 60-ft tree is drawn with 1 in. : 15 ft. How tall is the drawing?

UNIT 8 · LESSON 8.6 · PRACTICE

Lesson 8.6: Area in Scale Drawings

Name _____ Date _____

I can find actual areas from scale drawings.

Warm-Up

1. 200 × 300

2. 50 × 50

Find the actual area

3. Drawing 2 × 3 in.; 1 in. : 5 ft

4. Drawing 5 × 8 cm; 1 cm : 2 m

5. Drawing 3/2 × 4 in.; 1 in. : 6 ft

6. Drawing 5/2 × 7/2 cm; 1 cm : 10 m

Solve

7. A rectangular rug on a plan (1 in. : 2 ft) measures 2.5 in. by 4 in. Carpet costs $3 per ft². Cost of the rug?

Exit Ticket

8. A plan uses 1 cm : 3 m. A room is 4 cm by 5 cm on the plan. Find the actual area.

UNIT 8 · LESSON 8.7 · PRACTICE

Lesson 8.7: Maps and Scale

Name _____ Date _____

I can use a map scale to find real distances and travel times.

Warm-Up

1. 7.2 × 25

2. 300,000 cm in km

Find the actual distance

3. Scale 1 cm : 40 km; map 3.5 cm

4. Scale 1 in. : 15 mi; map 6 in.

5. Scale 1 : 25,000; map 8 cm

6. Scale 1 : 200,000; map 5 cm

Solve

7. On a map (1 in. : 50 mi), Ellie's trip is 4.8 in. She drives 60 mph. How long is the drive?

8. A hiking map is 1 : 24,000. A trail is 15 cm on the map. How many kilometers is it?

Exit Ticket

9. Two towns are 75 km apart. With 1 cm : 15 km, how far apart on the map?

UNIT 8 · LESSON 8.8 · PRACTICE

Lesson 8.8: Changing the Scale

Name _____ Date _____

I can reproduce a scale drawing at a different scale.

Warm-Up

1. 10 × 2

2. 20 ÷ 4

Redraw at the new scale

3. 6 in. at 1 in. : 4 ft; new scale 1 in. : 8 ft

4. 5 cm at 1 cm : 10 m; new scale 1 cm : 5 m

5. 9 in. at 1 in. : 2 ft; new scale 1 in. : 3 ft

6. 4 cm at 1 cm : 3 m; new scale 1 cm : 2 m

Exit Ticket

7. A rectangle 8 cm × 4 cm uses 1 cm : 3 m. Redraw it at 1 cm : 6 m on the grid (1 square = 1 cm).

WEEKLY CHECK · Fri Jun 25 · ABOUT 15 MINUTES

Weekly Check 23: Scale Drawings and Maps

Name _____ Date _____

Show what you know

1. Scale 1 in. : 8 ft; actual 36 ft. Drawing?

2. Drawing 3 cm × 4 cm; 1 cm : 5 m. Actual area?

3. Map 1 cm : 30 km; 4.5 cm. Actual?

4. Scale 1 : 50,000; map 4 cm. Actual km?

5. Drawing 10 cm at 1 cm : 2 m. Redraw at 1 cm : 5 m.

6. 180 mi at 45 mph: hours?

UNIT 8 · LESSON 8.9 · PRACTICE

Lesson 8.9: Triangle Side Lengths

Name _____ Date _____

I can decide whether three side lengths can form a triangle.

Warm-Up

1. 2 + 3

2. Is 5 > 6?

Triangle? Write yes or no.

3. 5, 6, 7

4. 2, 2, 5

5. 10, 4, 7

6. 3, 9, 6

7. 12, 5, 13

8. 1.5, 2.5, 3.5

Find the range

9. Two sides are 5 cm and 8 cm. What lengths could the third side be?

10. Two sides are 10 in. and 4 in. Can the third side be 6 in.? 7 in.?

Exit Ticket

11. Explain why 4, 5, 10 cannot form a triangle.

UNIT 8 · LESSON 8.10 · PRACTICE

Lesson 8.10: Triangle Angles

Name _____ Date _____

I can use the angle sum of a triangle to decide whether angles can form a triangle.

Warm-Up

1. 180 − 48 − 75

2. 90 + 60 + 40

Find the missing angle

3. 60° and 60°

4. 90° and 25°

5. 120° and 35°

6. 45° and 45°

7. 100° and 47°

8. 15° and 82°

Can these be the angles of a triangle?

9. 70°, 70°, 40°

10. 90°, 90°, 10°

11. 100°, 40°, 50°

12. 33°, 57°, 90°

Exit Ticket

13. A right triangle has one 38° angle. Find the third angle.

UNIT 8 · LESSON 8.11 · PRACTICE

Lesson 8.11: Drawing Triangles from Conditions

Name _____ Date _____

I can draw triangles with a ruler and protractor and tell whether the conditions give one, many, or no triangles.

Warm-Up

1. Is 5 + 6 > 7?

2. 50 + 60 + 70

One, many, or none?

3. sides 3, 4, 8 cm

4. sides 6, 8, 10 cm

5. angles 30°, 60°, 90°

6. sides 5 and 9 cm with a 70° angle between

7. angles 40° and 80° with a 6 cm side between

8. angles 95°, 95°, ?

Draw it

9. Use a ruler and protractor to draw a triangle with sides 6 cm and 4 cm and a 60° angle between them. Measure the third side.

Exit Ticket

10 Why do three angles not give a unique triangle?

UNIT 8 · LESSON 8.12 · PRACTICE

Lesson 8.12: Drawing Quadrilaterals with Conditions

Name _____ Date _____

I can draw quadrilaterals with given conditions and decide whether they are unique.

Warm-Up

1 Angles in a rectangle

2 Angle sum of a quadrilateral

One or many?

3 square, side 5 cm

4 rhombus, side 5 cm

5 rectangle, 4 cm by 7 cm

6 quadrilateral with sides 3, 4, 5, 6 cm

Draw it

7 Draw a parallelogram with sides 5 cm and 3 cm and a 60° angle. Is it unique?

Exit Ticket

8 A quadrilateral has angles 90°, 90°, 90°, and ? Find the fourth angle.

WEEKLY CHECK · Fri Jul 2 · ABOUT 15 MINUTES

Weekly Check 24: Triangles and Quadrilaterals

Name _____ Date _____

Show what you know

1. Triangle? 4, 6, 11

2. Triangle? 7, 8, 12

3. Missing angle: 64° and 71°

4. Angles 50°, 60°, 70°: one, many, or none?

5. Sides 5 cm, 7 cm with 45° between: one, many, or none?

6. Rhombus with side 4 cm: one or many?

UNIT 8 · LESSON 8.13 · PRACTICE

Lesson 8.13: Cross Sections of Prisms

Name _____ Date _____

I can describe the cross sections made by slicing a rectangular prism.

Warm-Up

1. How many faces does a cube have?

2. Shape of each face of a cube

Name the cross section

3. A box 3 × 5 × 8, sliced parallel to the 3 × 5 face

4. A cube sliced diagonally from one top edge to the opposite bottom edge

5. A loaf of bread sliced like usual

6. A triangular prism sliced parallel to its triangle base

Exit Ticket

7. Can a slice of a cube be a pentagon? Try to sketch one.

UNIT 8 · LESSON 8.14 · PRACTICE

Lesson 8.14: Cross Sections of Pyramids

Name _____ Date _____

I can describe the cross sections made by slicing a pyramid.

Warm-Up

1. Shape of a square pyramid's base

2. Shape of its side faces

Name the cross section

3. Square pyramid, slice parallel to the base near the top

4. Square pyramid, vertical slice through the apex

5. Rectangular pyramid, horizontal slice

6. Triangular pyramid, horizontal slice

Exit Ticket

7. A square pyramid's base is 10 in. on a side. A slice halfway up and parallel to the base: what shape and size?

UNIT 8 · LESSON 8.15 · PRACTICE

Lesson 8.15: Cross Sections of Cylinders and Cones

Name _____ Date _____

I can describe the cross sections of cylinders and cones.

Warm-Up

1. Shape of a cylinder's bases

2. Shape of a cone's base

Name the cross section

3. Cylinder, horizontal

4. Cylinder, vertical through center

5. Cone, horizontal

6. Cone, vertical through tip

7. Cylinder ($r = 3$, $h = 10$), vertical through center: dimensions

8. Sphere, any slice

Exit Ticket

9. A cylinder has diameter 4 in. and height 9 in. Describe its vertical cross section through the center.

UNIT 8 · LESSON 8.16 · PRACTICE

Lesson 8.16: Scale Drawing Project

Name _____ Date _____

I can create a scale drawing of a real space and use it to solve problems.

Warm-Up

1. 14 ÷ 2

2. 14 × 12

Project

3. Measure your room. Length and width:

4. Choose a scale:

5. Draw your room and at least three objects on the grid.

6. Find the floor area of your room.

Exit Ticket

7 Why did you choose your scale?

UNIT REVIEW · Thu Jul 8 (homework) or as needed · ABOUT 40 MINUTES

Unit 8 Review

Name _____ Date _____

Scale

1. Scale factor from 15 to 6

2. Area 20, scale factor 1.5

3. Plan 1 in. : 3 ft; room 5 in. × 4.5 in. Actual size?

4. Same room: actual area?

5. Map 1 : 100,000; 7.5 cm. Actual km?

6. Redraw 9 cm at 1 cm : 4 m using 1 cm : 6 m

Geometry

7. Two sides 6 and 9: possible third sides

8. Missing angle: 90° and 27°

9. ASA: one, many, or none?

10. Cross section of a cylinder, horizontal

11. Cross section of a square pyramid through the apex, vertical

12. Cross section of a 2 × 3 × 7 box parallel to the 2 × 3 face

UNIT TEST · Fri Jul 9 · ABOUT 45 MINUTES

Unit 8 Test

Name _____ Date _____

Show your work. This test has 21 points.

Part A · Scale (1 point each)

1 Is 9 × 12 a scaled copy of 3 × 4?

2 Scale factor from 10 to 25

3 Side 18 cm, scale factor $\frac{1}{3}$

4 Area 12, scale factor 2

5 Scale 1 in. : 6 ft; drawing 4.5 in. Actual?

6 Scale 1 cm : 25 km; actual 200 km. Drawing?

Part B · Triangles and figures (1 point each)

7 Triangle? 3, 5, 9

8 Missing angle: 38° and 106°

9 Sides 6, 8, 10: one, many, or none?

10 Angles 45°, 45°, 90°: one, many, or none?

11 Cone, horizontal slice

12 Cube, slice parallel to a face

Part C · Problem solving (3 points each)

13 A floor plan uses 1 in. : 4 ft. A living room is 5 in. by 3.5 in. on the plan. Find the actual dimensions and area. Carpet costs $2.50 per square foot: how much to carpet the room?

14 A map uses 1 cm : 20 km. Two towns are 9 cm apart. Ellie drives 60 km/h. How long is the drive? Show your work.

15 Use a ruler and protractor to draw a triangle with a 6 cm side between angles of 50° and 70°. Find the third angle. Is the triangle unique?

UNIT 9

Angles, Circles, Area, and Volume

Angle relationships, circles, composite area, surface area and volume of prisms, and a review of the whole year.

Your practice pages for Unit 9. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

supplementary	Sum to 180°.
complementary	Sum to 90°.
circumference	$C = \pi d$.
area of a circle	$A = \pi r^2$.
volume of a prism	$V = Bh$.

UNIT 9 · LESSON 9.1 · PRACTICE

Lesson 9.1: Angle Relationships

Name _____ Date _____

I can identify complementary, supplementary, vertical, and adjacent angles.

Warm-Up

1. 90 − 35

2. 180 − 130

Find the angle

3. Complement of 40°

4. Supplement of 40°

5. Complement of 72°

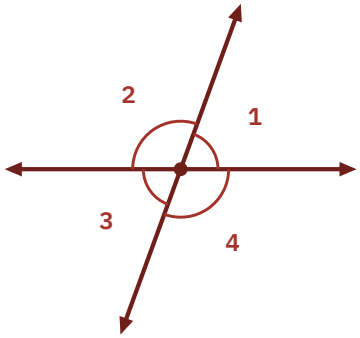
6. Supplement of 115°

7. Vertical angle to 33°

8. Neighbor (adjacent, on a line) of 33°

Name the relationship

9.



10. Can two angles be both complementary and vertical? Explain.

Exit Ticket

11. Two angles are supplementary. One is 4 times the other. Find both.

UNIT 9 · LESSON 9.2 · PRACTICE

Lesson 9.2: Solving for Unknown Angles

Name _____ Date _____

I can write and solve equations to find unknown angles.

Warm-Up

1. $2x + 80 = 180$

2. $3x = 105$

Write an equation and solve

3. Complementary: $x°$ and $64°$

4. Supplementary: $(x + 30)°$ and $2x°$

5. Vertical: $(5x - 10)°$ and $80°$

6. Complementary: $(3x)°$ and $(x + 10)°$

7. Three angles on a line: $40°$, $x°$, and $2x°$

8. Angles around a point: $90°$, $150°$, and $x°$

Exit Ticket

9. Two vertical angles measure $(6x + 4)°$ and $(4x + 20)°$. Find x and the angle.

UNIT 9 · LESSON 9.3 · PRACTICE

Lesson 9.3: Circles and Circumference

Name _____ Date _____

I can find the circumference of a circle from its radius or diameter.

Warm-Up

1. 3.14 × 10

2. 2 × 7

Find the circumference

3. d = 4 cm

4. r = 5 in.

5. d = 20 ft

6. r = 12 m

7. d = 5/2 cm

8. r = 50 mm

Solve

9. A bike wheel has diameter 26 in. How far does it roll in one turn? In 100 turns?

10. A circular track is 314 m around. What is its diameter?

Exit Ticket

11. A circle has radius 9 cm. Find its circumference.

UNIT 9 · LESSON 9.4 · PRACTICE

Lesson 9.4: Area of a Circle

Name _____ Date _____

I can find the area of a circle.

Warm-Up

1. 5^2

2. 3.14×25

Find the area

3. r = 2 cm

4. r = 7 in.

5. d = 10 ft

6. d = 12 m

7. r = 20 yd

8. d = 3 cm

Solve

9. A sprinkler waters a circle of radius 15 ft. What area does it water?

10. A 12-in. pizza (diameter) costs $12. A 16-in. pizza costs $18. Which is the better deal per square inch?

Exit Ticket

11. Find the area of a circle with diameter 14 cm.

WEEKLY CHECK · Fri Jul 16 · ABOUT 15 MINUTES

Weekly Check 25: Angles and Circles

Name _____ Date _____

Show what you know

1. Complement of 37°

2. Vertical angles (3x + 6)° and 51°: x?

3. Supplementary: x° and (2x + 30)°

4. Circumference, r = 5 in.

5. Area, d = 10 cm

6. Diameter if C = 47.1 m

UNIT 9 · LESSON 9.5 · PRACTICE

Lesson 9.5: Problems with Circles

Name _____ Date _____

I can find areas and perimeters of figures made from parts of circles.

Warm-Up

1. 3.14 × 25 ÷ 2

2. 3.14 × 10 ÷ 2

Find the area

3. Semicircle, r = 6 m

4. Quarter circle, r = 2 cm

5. Square 8 × 8 with a circle (d = 8) cut out

6. Rectangle 12 × 4 with semicircles on both short ends

Solve

7. A running track is a 100 m by 60 m rectangle with semicircles on both 60 m ends. Find the distance around the track.

Exit Ticket

8. Find the area of the window above if the rectangle is 4 ft by 3 ft (semicircle on the 4 ft side).

UNIT 9 · LESSON 9.6 · PRACTICE

Lesson 9.6: Area of Composite Figures

Name _____ Date _____

I can find the area of composite figures made of triangles, rectangles, and trapezoids.

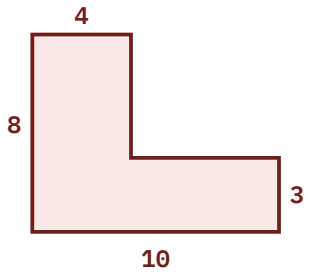
Warm-Up

1. $12 \times 4 + 5 \times 5$

2. $(10 + 5) \div 2 \times 4$

Find the area

3. L-shape: full width 10, full height 8, top width 4, bottom height 3



4. Trapezoid: bases 6 and 14, height 5

5. A rectangle 8 × 5 with a triangle (base 8, height 3) on top

6. Parallelogram b = 12, h = 7

Solve

7. A deck is a 16 ft by 12 ft rectangle with a 4 ft by 4 ft square cut out for a tree. Stain covers 50 ft² per can. How many cans?

Exit Ticket

8. Find the area: a 10 × 6 rectangle with a right triangle (legs 6 and 4) attached to the right side.

UNIT 9 · LESSON 9.7 · PRACTICE

Lesson 9.7: Surface Area of Prisms

Name _____ Date _____

I can find the surface area of rectangular and triangular prisms.

Warm-Up

1. $2 \times (15 + 20 + 12)$

2. $\frac{1}{2} \times 6 \times 8$

Find the surface area

3. Box $4 \times 4 \times 4$

4. Box $8 \times 3 \times 2$

5. Box $10 \times 5 \times 6$

6. Box $3/2 \times 2 \times 4$

7. Triangular prism: legs 3 and 4, hypotenuse 5, length 10

8. Triangular prism: legs 5 and 12, hypotenuse 13, length 8

Solve

9. Lillie wraps a gift box 12 in. × 8 in. × 3 in. How much wrapping paper covers it exactly?

Exit Ticket

10. Find the surface area of a box 7 cm × 2 cm × 5 cm.

UNIT 9 · LESSON 9.8 · PRACTICE

Lesson 9.8: Volume of Prisms

Name _____ Date _____

I can find the volume of any right prism using $V = Bh$.

Warm-Up

1. $5 \times 3 \times 4$

2. 24×12

Find the volume

3. Box $9 \times 2 \times 3$

4. Box $5/2 \times 4 \times 6$

5. Box $10 \times 10 \times 1/2$

6. Triangular prism: legs 3 and 4, length 10

7. Trapezoid base (bases 5, 9; height 2), length 6

8. $B = 32.5$ in², $h = 4$ in.

Solve

9. A fish tank is 30 in. × 12 in. × 16 in. One gallon is about 231 in³. About how many gallons does it hold?

Exit Ticket

10. A prism's base is a triangle with base 10 cm and height 6 cm. The prism is 9 cm long. Find its volume.

WEEKLY CHECK · Fri Jul 23 · ABOUT 15 MINUTES

Weekly Check 26: Area, Surface Area, and Volume

Name _____ Date _____

Show what you know

1. Semicircle area, r = 10

2. Trapezoid: bases 8 and 12, height 5

3. Surface area, box 4 × 3 × 2

4. Volume, box 4 × 3 × 2

5. Volume, triangular prism: legs 5, 6; length 10

6. SA, cube edge 5

UNIT 9 · LESSON 9.9 · PRACTICE

Lesson 9.9: Volume and Surface Area in Context

Name _____ Date _____

I can decide whether a problem needs volume or surface area and solve it.

Warm-Up

1. $6 \times 4 \times 1$

2. $2 \times 6 + 2 \times 4$

Volume or surface area? Then solve.

3. How much soil fills a planter 3 ft × 2 ft × 1.5 ft?

4. How much fabric covers a cube-shaped ottoman, 2 ft edges, all faces except the bottom?

5. An open-top box (no lid) is 10 cm × 8 cm × 5 cm. How much cardboard?

6. Ice cubes are 2 cm on each edge. How many fit in a 10 cm × 8 cm × 6 cm tray (exactly packed)?

Exit Ticket

7. Paint covers 350 ft² per gallon. Walls of a room total 1,050 ft². How many gallons?

UNIT 9 · LESSON 9.10 · PRACTICE

Lesson 9.10: Composite Solids

Name _____ Date _____

I can find the volume and surface area of solids made from prisms.

Warm-Up

1. $6 \times 3 \times 1 + 3 \times 3 \times 1$

2. $1{,}000 - 40$

Find the volume

3. L-shaped block: $10 \times 4 \times 2$ plus $4 \times 4 \times 2$

4. Box $12 \times 6 \times 6$ minus a cube of edge 3

5. Triangular prism (legs 4, 6; length 10) on a box $6 \times 10 \times 2$

6. Three stacked boxes: $6 \times 6 \times 1$, $4 \times 4 \times 1$, $2 \times 2 \times 1$

Exit Ticket

7. A planter is a 4 ft × 2 ft × 2 ft box with a 4 ft × 1 ft × 1 ft box attached on the side. How much soil does it hold?

UNIT 9 · LESSON 9.11 · PRACTICE

Lesson 9.11: Comparing Volume and Surface Area

Name _____ Date _____

I can compare solids with the same volume but different surface areas.

Warm-Up

1. $8 \times 1 \times 1$

2. $2 \times 2 \times 2$

Volume 24: find the surface area

3. $24 \times 1 \times 1$

4. $12 \times 2 \times 1$

5. $6 \times 4 \times 1$

6. $4 \times 3 \times 2$

Explain

7. A company wants a box that holds 24 cubes and uses the least cardboard. Which box above should it choose? Why?

Exit Ticket

8. Can two boxes have the same surface area but different volumes? Give an example.

UNIT 9 · LESSON 9.12 · PRACTICE

Lesson 9.12: Geometry Problem Solving

Name _____ Date _____

I can choose and use the right geometry formulas to solve multi-step problems.

Warm-Up

1 13^2

2 $169 - 100$

Solve

3 A circular table has diameter 4 ft. A cloth hangs 1 ft over the edge all around. What is the cloth's area?

4 Two angles are supplementary. One is 24° more than the other. Find both.

5 A tent is a triangular prism: triangle base 6 ft, height 4 ft, length 8 ft. Find the space inside.

6 A cylindrical-looking hay bale is wrapped with a band around its 5 ft diameter. How long is the band?

Exit Ticket

7 Write your own two-step geometry problem and solve it.

WEEKLY CHECK · Fri Jul 30 · ABOUT 15 MINUTES

Weekly Check 27: Geometry Problem Solving

Name _____ Date _____

Show what you know

1. Filling a tank: volume or SA?

2. Open box 5 × 4 × 3: cardboard

3. Box 6 × 6 × 6 minus a 2 × 2 × 6 hole

4. Which has less SA: 6 × 1 × 1 or 3 × 2 × 1?

5. Ring: outer r = 5, inner r = 3

6. Two complementary angles; one is twice the other

UNIT 9 · LESSON 9.13 · PRACTICE

Lesson 9.13: Year Review: Ratios, Proportions, and Percents

Name _____ Date _____

I can solve problems with unit rates, proportional relationships, and percents.

Warm-Up

1. 4.47 ÷ 3

2. 7.25 ÷ 5

Review: Units 1–2

3. Ellie reads 45 pages in 1.5 hours. Pages per hour?

4. Solve: $\frac{6}{15} = \frac{x}{40}$

5. Is y = 4x + 1 proportional?

6. 15% tip on $48

7. Price $120 → $90: percent change

8. Simple interest: $500 at 4% for 3 years

9. 6% tax on $35

10. Map: 2 cm : 15 km. 7 cm?

Exit Ticket

11. A recipe uses 3 cups flour for 2 dozen cookies. How much for 5 dozen?

UNIT 9 · LESSON 9.14 · PRACTICE

Lesson 9.14: Year Review: Rational Numbers

Name _____ Date _____

I can add, subtract, multiply, and divide rational numbers.

Warm-Up

1. $-6 + 14$

2. $8 - 9$

Review: Units 3–5

3. $-15 + (-9)$

4. $4 - 11$

5. $-1.2 + 3.5$

6. $(-6)(-7)$

7. $-48 \div 6$

8. $\frac{2}{3} \times (-\frac{9}{10})$

9. Write $\frac{5}{8}$ as a decimal

10. Write $\frac{2}{3}$ as a decimal

11. $(-2)^3$

12. $|-9| - |4|$

Exit Ticket

13. A diver at −40 ft rises 15 ft, then descends 28 ft. Where is she?

UNIT 9 · LESSON 9.15 · PRACTICE

Lesson 9.15: Year Review: Expressions, Equations, and Inequalities

Name _____ Date _____

I can simplify expressions and solve equations and inequalities.

Warm-Up

1. $110 - 20$

2. $90 \div 15$

Review: Unit 6

3. Simplify $5a - 2(a + 3)$

4. Simplify $\frac{1}{2}(8x - 6) + x$

5. Evaluate $2x^2 - x$ when $x = -3$

6. $-2(x + 3) = 10$

7. $\frac{1}{2}x - 2 = 4$

8. $0.4n + 1.2 = 3.6$

9. $-4y \geq 20$

10. $2x - 9 < 7$

Exit Ticket

11. Write and solve: three times a number, minus 8, is 25.

UNIT 9 · LESSON 9.16 · PRACTICE

Lesson 9.16: Year Review: Statistics, Probability, and Geometry

Name _____ Date _____

I can solve problems using statistics, probability, and geometry.

Warm-Up

1. 0.35 × 600

2. 6 × 6

Review: Units 7–9

3. Mean and MAD of 4, 6, 8, 10, 12

4. Bag: 3 red, 5 blue, 2 green. P(not blue)

5. Outfits: 4 shirts, 3 pants

6. Scale 1 in. : 6 ft. Drawing 3.5 in.?

7. Triangle? 3, 4, 8

8. Supplement of 47°

9. Circumference, d = 15 cm

10. Volume, box 6 × 4 × 2.5

11. Surface area, box 6 × 4 × 2

12. Cross section of a cylinder, horizontal

Exit Ticket

13. Which topic from this year do you feel strongest in? Which one will you review before the test?

YEAR-END REVIEW · Thu Aug 5 (homework) or as needed · ABOUT 50 MINUTES

Year-End Review

Name _____ Date _____

Ratios, percents, and rational numbers

1. Unit price: 6 for $8.70

2. y = kx; y = 21 when x = 3. Find y when x = 8.

3. 35% of 240

4. Price $60 → $75: percent change

5. −8 − (−3) + 12

6. $-\frac{5}{6} \times \frac{3}{10}$

7. −3.6 ÷ (−0.4)

8. $\frac{7}{20}$ as a decimal

Algebra

9. Simplify 4(2x − 1) − 3x

10. −5x + 8 = −17

11. 3(n − 2) = 15

12. 6 − 2x ≥ 14

Statistics, probability, and geometry

13. 18 of 60 sampled like a show; estimate for 2,500

14. Coin + cube: P(T and even)

15. Area, circle r = 4

16. Volume, triangular prism: legs 4 and 9, length 5

17 Scale 1 cm : 8 m; 6.5 cm

18 Missing angle: 90° and 34° in a triangle

YEAR-END TEST · Fri Aug 6 · ABOUT 60 MINUTES

Grade 7 Year-End Test

Name _____ Date _____

Show your work. This test has 32 points.

Part A · Ratios and percents (1 point each)

1. Unit rate: 150 miles in 2.5 hours

2. $y = kx$ passes through (5, 35). Find k.

3. 30% of 90

4. A $40 shirt is 15% off. Sale price?

5. Price $50 → $65: percent change

Part B · Rational numbers (1 point each)

6. $-14 + 9$

7. $6 - (-11)$

8. $(-4)(-2.5)$

9. $-\frac{2}{3} \div \frac{4}{9}$

10. Write $\frac{3}{8}$ as a decimal

Part C · Expressions, equations, inequalities (1 point each)

11. Simplify $2(3x - 5) + 4x$

12. Factor $24n - 16$

13. $-6x + 4 = 40$

14. $5(y + 2) = -15$

15 $-2k + 7 \leq 15$

Part D · Statistics and probability (1 point each)

16 A random sample of 50 has 12 left-handed people. Estimate for 1,500.

17 Mean of 3, 7, 8, 10, 12

18 Bag: 4 red, 6 blue. P(blue)

19 Two coins: P(at least one head)

20 Two number cubes: P(sum is 10)

Part E · Geometry (1 point each)

21 Scale 1 in. : 12 ft; 3.5 in. Actual?

22 Supplement of 68°

23 Circumference, d = 12 m

24 Area, r = 9 cm

25 Volume, box 7 × 3 × 4

26 Surface area, box 7 × 3 × 4

Part F · Problem solving (3 points each)

27 A phone plan costs $25 per month plus $0.10 per text. Ellie can spend at most $40 per month. Write and solve an inequality for the number of texts. Interpret.

28 A circular garden has diameter 14 ft. Find its area. Edging costs $2 per foot: how much to edge it? Show your work.

Grade 7 Mathematics

Student Workbook · 2026–2027

A full year of Grade 7 mathematics, taught one clear idea at a time. Every lesson opens with a question worth thinking about, walks through worked examples step by step, and builds from models and pictures to fluent, confident problem solving.

This Student Workbook holds the daily practice pages for every lesson, plus each Friday's Weekly Check, the unit reviews, and the unit tests, with room to show thinking and write complete answers.

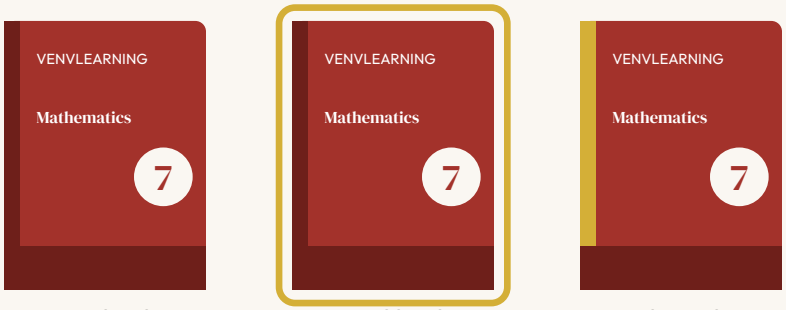
INSIDE THIS YEAR

01 Proportional Relationships
02 Percents
03 Adding and Subtracting Integers
04 Adding and Subtracting Rational Numbers
05 Multiplying and Dividing Rational Numbers
06 Expressions, Equations, and Inequalities
07 Statistics and Probability
08 Scale Drawings and Geometric Figures
09 Angles, Circles, Area, and Volume

IN THIS BOOK

- Warm-ups, practice, and exit tickets for every lesson
- Weekly Checks every Friday
- Unit reviews and unit tests
- Space to show work and explain thinking

THE COMPLETE SET



Textbook

Workbook

You are holding this book

Teacher Edition

Each lesson has the same number in all three books, and every day of the year is mapped to its lesson in the VenvLearning planner.